Supplementary material: Masking cannot remove what is inside the region of interest

Contents and map

Each section below supports the main-text section named in the third column; the main text cites them as "Supplement Sn". Every interval in this supplement is a crossed seed×image bootstrap interval unless the section says otherwise (main text, Section 4).

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S1. Pre-registration index and registry

Every experiment was specified in a dated document committed to the public repository before any model was fitted on the corresponding data; amendments were committed before the amended analysis. The registry lists, for every registration, its first commit and the first commit that added outcome results (outcome-free cell counts excluded; for registrations that add arms to an existing results folder, the first commit whose results contain the new arm). Commit times are set by the committing machine and are therefore not an external time stamp; the registrations of the second review round (`PREREGISTRATION_REVIEW2`) were additionally pushed to GitHub before any model was fitted. The primary family used for multiplicity control (`STATISTICAL_PLAN`) was assembled after the individual results were available; each member test was individually pre-registered, and the Holm correction is applied to the whole family. Generated by `scripts/verify/registry_table.py`.

Registration	Registered (commit, UTC)	First outcome result (commit, UTC)	Order
PREREGISTRATION_ISIC2019_TRAPS	76e25d4 2026-09-26 13:14	2699d51 2026-09-26 15:47	registration first
PRECOMMIT_MATCHED_TRAPS	5ef3228 2026-09-26 14:57	2699d51 2026-09-26 15:47	registration first
PREREGISTRATION_CXR_DEVICE_TRAPS	4d1a924 2026-09-26 18:07	e0d5310 2026-09-26 18:44	registration first
PREREGISTRATION_UNIVERSAL_I2E	ee95f10 2026-09-26 20:14	2f09c07 2026-09-26 21:07	registration first
PREREGISTRATION_THYROID_TRAPS	ea57661 2026-09-26 22:46	b7bc5e7 2026-09-27 00:07	registration first
PREREGISTRATION_SLAS	2d1ba20 2026-09-26 23:33	2d1ba20 2026-09-26 23:33	same commit
PREREGISTRATION_CAPSULE_TRAPS	fcbf8d9 2026-09-27 00:40	6c3e8c5 2026-09-27 00:59	registration first
PREREGISTRATION_CXR_DRAIN	3d0cffc 2026-09-27 03:29	1ef04b7 2026-09-27 04:16	registration first
PREREGISTRATION_FINETUNE	7c8b466 2026-09-27 07:06	1f4f6e3 2026-09-27 08:00	registration first
PREREGISTRATION_OVARY_TRAPS	2517600 2026-09-27 07:29	cd67b8e 2026-09-27 07:45	registration first
STATISTICAL_PLAN	4e9d44c 2026-09-27 09:37	4e9d44c 2026-09-27 09:37	same commit
PREREGISTRATION_NATURAL	74bd243 2026-09-27 09:39	be95223 2026-09-27 09:50	registration first
PREREGISTRATION_INPAINT_LAMA	715b790 2026-09-27 09:41	ae374fd 2026-09-27 10:14	registration first
PREREGISTRATION_TEXT_PROMPT	7877e8c 2026-09-27 12:21	2203aa2 2026-09-27 12:54	registration first
PREREGISTRATION_THEORY_PREDICTION	4f78653 2026-09-27 13:46	30ce70a 2026-09-27 14:03	registration first
PREREGISTRATION_EMBEDDING_GROUPS	b0eda23 2026-09-27 13:55	30ce70a 2026-09-27 14:03	registration first
PREREGISTRATION_SCALE	13303b0 2026-09-27 14:13	cef4871 2026-09-27 14:49	registration first
PREREGISTRATION_UMTE_ABLATION	30f19c8 2026-09-27 14:15	0abd4a7 2026-09-27 14:22	registration first
PREREGISTRATION_FT_ERASE	8ac7a7d 2026-09-27 14:20	3d0dd70 2026-09-27 14:36	registration first
PREREGISTRATION_FT_UMTE	b7d7077 2026-09-27 14:52	f28ab78 2026-09-27 15:27	registration first
PREREGISTRATION_REVIEW2	29cd0da 2026-09-27 18:56	3673625 2026-09-27 19:40	registration first
PREREGISTRATION_REVIEW3	85da735 2026-09-27 23:43	e4ffc8a 2026-09-28 00:11	registration first
PREREGISTRATION_ROUND4	d34b8b8 2026-09-28 11:28	ac90660 2026-09-28 12:48	registration first
PREREGISTRATION_ROUND5	4af106a 2026-09-28 13:37	a02db23 2026-09-28 14:01	registration first
PREREGISTRATION_ROUND6	607900c 2026-09-28 14:27	4c3263d 2026-09-28 16:29	registration first
PREREGISTRATION_ROUND7	cbf4229 2026-09-28 20:37	68e2432 2026-09-28 20:49	registration first
PREREGISTRATION_ROUND8	dd5c7ee 2026-09-28 22:01	0d66e42 2026-09-29 00:12	registration first
PREREGISTRATION_ROUND9	6563a66 2026-09-29 08:44	76739e3 2026-09-29 09:19	registration first

Registration status and multiplicity of every analysis family.

Analysis	Registration status	Multiplicity
Real traps, controlled sweeps, natural data, U-MtE (P1–P4)	before results	16-test primary family (Holm)
Primary-family grouping	after results (post hoc)	—
Rebuild on new hardware (R0)	before results	tolerance criterion
Covariate-matched traps (R1)	designed after the real-trap results; registered before its own results	own family (Holm, 4)
Real-artifact transplant (R2)	designed after the real-trap results; registered before its own results	own family (Holm, 4)
Operating points (R3); conflicting subgroup	test registered before computing; subgroup defined post hoc	decision rule
Fine-tuned hair, ovary power (R4)	before results	single tests
Paste-edge control (R5)	designed after the transplant results; registered before its own results	own family (Holm, 4)
Crossed bootstrap (R6), stricter matching (R7)	designed after the results; registered before computing	re-estimation / sensitivity

S2. Replication of the original pilot

Of 88 numbers reported in the thesis proposal, 73 are reproduced exactly (same sign, same confidence-interval verdict, $|\Delta| \leq 0.02$), 12 agree in sign and significance with a larger difference, and 3 differ (a borderline occlusion interval, a contrast trap that was reported as null and is not, and one source stratum). Details: `results/SUMMARY.md`. The archived pilot confidence intervals are recomputed exactly from the archived predictions (`make verify`).

S3. Datasets considered and not used

CANDID-PTX (not accessible outside Health New Zealand); RANZCR-CLiP devices linked to NIH, kept as a boundary rather than a confirmation (Table 1: with the primary radiograph encoder RAD-DINO masking helped at both locations; with the registered MedSigLIP replication, fitted for the first time when these runs were regenerated, the crossover is positive for all four findings; device placement is confounded with other devices; the registered DINOv2 replication was not run); chest-radiograph cardiomegaly with lines over the heart (210 labelled images); TCGA/GrandQC pathology (pen-mark masks unreliable); ISIC ink (70 in-lesion images); breast ultrasound BUSI/BUS-BRA (caliper-free group only $\approx 70\%$ pure after a two-detector design); DDTI thyroid (TI-RADS instead of pathology; 134 images); BKAI-IGH colonoscopy specular highlights (no highlight-free images). *External validation with patient identifiers.* TN-SCUI 2020 (one image per patient, nodule masks) may not be used outside the challenge under its terms. ThyUS2Path [7] (842 patients, 8,508 images, surgical pathology, CC BY 4.0) contains real calipers in several colours but no nodule masks, and our frozen caliper detector responds to elements of its scanner interface (frame lines, depth scales) on a random 25% sample inspected before any label was read. Using it requires a nodule

segmentation transferred from TN3K and a caliper detector re-validated on this interface; we did not run it.

Table 1: Chest radiographs with real devices (NIH ChestX-ray14 linked to RANZCR-CLiP, AP films): Trap A = central venous catheter inside the lungs, Trap B = endotracheal tube outside them; reversed-test AUROC, crossed 95% CIs, Holm over the four findings per encoder. Registered claims: X1 masking helps in Trap B, X2 masking harms in Trap A, X3 crossover > 0. RAD-DINO is the registered primary encoder; the registered MedSigLIP replication was fitted for the first time when these runs were regenerated.

Encoder	Finding	mask − ERM, Trap A (in lungs)	mask − ERM, Trap B (outside)	Crossover	p (Holm, 4)	Supported
RAD-DINO	Atelectasis	$+0.059$ $[+0.019, +0.099]$	$+0.105$ $[+0.092, +0.117]$	$+0.046$ $[+0.011, +0.082]$	0.0544	X1
	Consolidation	$+0.074$ $[+0.025, +0.125]$	$+0.070$ $[+0.047, +0.092]$	-0.004 $[-0.065, +0.058]$	0.9112	X1
	Effusion	$+0.078$ $[+0.032, +0.125]$	$+0.100$ $[+0.089, +0.111]$	$+0.022$ $[-0.030, +0.074]$	0.8364	X1
	Infiltration	$+0.087$ $[+0.060, +0.116]$	$+0.063$ $[+0.055, +0.071]$	-0.025 $[-0.055, +0.006]$	0.3444	X1
MedSigLIP	Atelectasis	$+0.183$ $[+0.146, +0.220]$	$+0.278$ $[+0.261, +0.296]$	$+0.095$ $[+0.053, +0.137]$	0.0004	X1 X3
	Consolidation	$+0.087$ $[+0.031, +0.142]$	$+0.226$ $[+0.206, +0.246]$	$+0.139$ $[+0.081, +0.197]$	0.0004	X1 X3
	Effusion	-0.084 $[-0.125, -0.043]$	$+0.244$ $[+0.230, +0.259]$	$+0.328$ $[+0.287, +0.369]$	0.0004	X1 X2 X3
	Infiltration	$+0.174$ $[+0.128, +0.219]$	$+0.309$ $[+0.293, +0.325]$	$+0.135$ $[+0.084, +0.185]$	0.0004	X1 X3

Chest radiography in detail. With a synthetic tube swept into the lung fields of NIH pneumothorax radiographs (RAD-DINO), the encoder does not use an extra-pulmonary tube at all (reversed 0.913 versus clean 0.915 at no overlap), so masking has nothing to remove. On real devices (RANZCR-CLiP linked to NIH, four findings; Table 1) lung masking helped at both locations with RAD-DINO, the registered primary encoder (no crossover survives Holm correction over the four findings). With MedSigLIP, the registered replication encoder, which does use the out-of-lung tube, the crossover is positive for all four findings ($+0.095$ to $+0.328$, Holm $p < 0.001$) and masking harms inside the lungs for effusion (-0.084 $[-0.125, -0.043]$). The device contrast is confounded, however: images without an in-lung catheter carry more out-of-lung tubes, which lung masking removes. Real chest drains carry the shortcut mostly through correlates (the treated lung; lung masking gains $+0.039$ $[+0.019, +0.058]$ with RAD-DINO and $+0.005$ $[-0.013, +0.023]$ with DINOv2). With the evidence split by encoder and a confounded contrast, we report chest radiography as the boundary of the law, not as a fifth confirmation.

S4. Artifact labels: provenance, audits, sensitivity and an expert audit kit

Which labels are human and which are automatic.

Cohort	Human annotation	Automatic annotation (model or rules)
ISIC 2019 (hair)	diagnoses (ISIC); lesion masks for the 10,015 HAM10000 images (manual)	lesion masks for the other 15,316 images (U-Net trained on manual ISIC 2018 masks); hair/ruler masks (published semi-automatic masks)
TN3K/TNCD (thyroid)	nodule masks; benign/malignant from cytological biopsy	caliper masks for all 3,493 images (rule-based detector)
MMOTU (ovary)	tumour masks; tumour categories	caliper masks for all 1,202 images (same detector)
SEE-AI (capsule)	lesion boxes and classes; contamination masks for 22 frames (expert)	contamination masks for 5,459 of 5,481 frames (linear probe on DINOv2 patch tokens, IoU 0.62)
NIH ChestX-ray14	drain labels of pneumothorax images (NEATX, non-expert annotators)	disease labels (text-mined from reports); lung masks (TorchXRayVision); drain labels of pneumothorax-negative images (drain detector)
All cohorts	—	visual audits of the automatic labels (by the analysis team, which includes an AI agent; no clinician)

Visual audit of 40 random images per cell by the analysis team (not clinicians): thyroid detections 95% on real calipers; caliper-free thyroid images 95–97% without visible calipers; ovary 40/40 detections are overlays, 31/40 calipers; caliper-free ovary images 92% without calipers but ≈28% with other overlays. Capsule debris probe: pixel accuracy 0.92, IoU 0.62 against expert masks. Re-running with large-marker (≥ 50 px) and strict-location ($r \geq 0.7$ / $r < 0.05$) definitions (thyroid, ovary) and a strict capsule Trap B leaves every primary conclusion unchanged (`SENSITIVITY_ARTIFACT_LABELS`). *Expert audit (prepared, not yet carried out).* `scripts/audit/make_expert_audit_kit.py` draws 60 images per cohort (half flagged by the automatic label), renders each blind (ROI contour only) and with the automatic mask, and writes a rating sheet (artifact present; location relative to the ROI; mask quality); `scripts/audit/score_expert_audit.py` computes Cohen's κ between two independent raters and the precision and recall of the automatic labels against their consensus, with bootstrap CIs.

S5. Secondary method analyses

The four analyses below come from archived runs whose per-image predictions were not kept; their numbers are point estimates without intervals.

- *Patch-token suppression (SLAS).* A linear probe on DINOv2 patch tokens localises real hair from 5 annotated images (patch AUROC 0.93; 0.97 with 50), but dropping flagged tokens improves reversed AUROC only modestly (+0.02 hair, +0.08 thyroid over ROI pooling).
- *Annotation-free pseudo-groups.* An overlay detector trained on synthetic insertion pairs recognises real calipers (AUROC 0.84) but not hair (0.57); pseudo-group balancing adds +0.04 on thyroid only.
- *Erasure combined with DFR.* Adds ≤ 0.014 over masking plus DFR.

- *Split-validation selector.* Admitting DFR as a candidate (trained and scored on disjoint validation halves) fixes the chest-drain case (0.656 vs 0.538) but makes choices noisier elsewhere; neither selector is uniformly best.

S6. Leakage

Holding cohort, features and protocol fixed, replacing the grouped split with random image-level splits (93 near-duplicate pairs across train/test) inflates the measured shortcut gap by 18% (0% overlap) and 10% (100%) and distorts both masking effects ($+0.184 \to +0.223$; $-0.129 \to -0.088$).

S7. Backbone scale

Replacing DINOv2 ViT-B/14 (86 M parameters) by ViT-S/14 (21 M) or ViT-L/14 (304 M), with everything else fixed (pre-registered), the location crossover is positive in all nine cohort–size combinations (ViT-S / ViT-B / ViT-L: thyroid +0.247 / +0.239 / +0.263; ovary +0.296 / +0.170 / +0.207; capsule +0.370 / +0.377 / +0.417; every CI excludes zero), and protected U-MtE beats masking on thyroid and ovary at every size (ViT-S +0.225 $[+0.168, +0.276]$ and +0.103 $[+0.074, +0.134]$; ViT-L +0.274 $[+0.243, +0.306]$ and +0.060 $[+0.037, +0.084]$). Larger models exploit the in-ROI shortcut *more*: the ERM correlated–reversed gap grows from 0.537 to 0.622 (thyroid) and from 0.320 to 0.564 (ovary) between ViT-S and ViT-L, so scale does not remove it. On capsule, disease protection is less complete with ViT-L (-0.044 $[-0.056, -0.032]$ versus masking).

S8. Mitigation methods and their results on the traps

The main text summarises the mitigations as a decision guide. This section gives the annotation-free mask-then-erase method (U-MtE), its results on the traps, the baselines and the end-to-end variants. None of these remedies is claimed to improve performance on unaltered test data (main text).

S8.1 Method: mask-then-erase with a generic insertion library

Insertion pairs. Removing an in-ROI artifact requires its pixels; inserting one does not. Given a frozen encoder f and an overlay generator $\mathcal{G}$, every training image x_i yields a counterfactual $x_i^+ = \mathcal{G}(x_i; \xi_i)$ with random parameters ξ_i and random position, half of the elements centred inside the ROI. Because insertion ignores the label, the differences $\Delta_i = f(x_i^+) - f(x_i)$ are independent of Y.

Generic library. $\mathcal{G}$ is one procedural family used unchanged for every modality and artifact: thin curvilinear strokes, thick bands, solid or translucent patches, blobs, dashed lines and small glyphs, with random colour (half greyscale), opacity and width. It contains no ruler, hair, caliper or debris template; it is a prior over "foreign overlays", not a model of any particular artifact.

Table 2: In-ROI artifacts (Trap A): reversed-test AUROC, with min(reversed, correlated) in parentheses (penalises shortcut flipping). Best robust value per column in bold. Generated by `scripts/make_main_table.py`.

Method	Needs	Hair/DINOv2	Hair/DermLIP	Thyroid/DINOv2	Thyroid/MedSigLIP	Thyroid/ConvNeXt	Ovary/DINOv2	Capsule/DINOv2	Capsule/MedSigLIP	Capsule/ConvNeXt
ERM	—	0.51 (0.51)	0.67 (0.67)	0.29 (0.29)	0.29 (0.29)	0.36 (0.36)	0.41 (0.41)	0.58 (0.58)	0.57 (0.57)	0.53 (0.53)
ROI masking	ROI	0.46 (0.46)	0.56 (0.56)	0.40 (0.40)	0.34 (0.34)	0.35 (0.35)	0.55 (0.55)	0.67 (0.67)	0.69 (0.69)	0.61 (0.61)
Mask + overlay augmentation	ROI	0.47 (0.47)	–	0.41 (0.41)	0.36 (0.36)	0.36 (0.36)	0.53 (0.53)	0.68 (0.68)	0.71 (0.71)	0.63 (0.63)
U-MtE (ours)	ROI	0.50 (0.50)	0.50 (0.50)	0.62 (0.62)	0.59 (0.59)	0.49 (0.49)	0.56 (0.56)	0.54 (0.54)	0.46 (0.46)	0.54 (0.54)
U-MtE, protected (ours)	ROI + Y of A=0	0.50 (0.50)	0.59 (0.59)	0.68 (0.68)	0.65 (0.65)	0.54 (0.54)	0.59 (0.59)	0.66 (0.66)	0.66 (0.66)	0.64 (0.64)
Group-balanced	A	**0.72 (0.72)**	**0.81 (0.81)**	0.68 (0.68)	0.70 (0.70)	0.68 (0.68)	0.70 (0.70)	0.76 (0.76)	0.80 (0.80)	0.72 (0.72)
DFR	A	0.71 (0.71)	0.78 (0.78)	0.75 (0.67)	0.75 (0.72)	0.73 (0.67)	0.72 (0.69)	0.82 (0.82)	0.82 (0.82)	0.78 (0.78)
Mask + DFR	ROI + A	0.66 (0.66)	–	0.83 (0.75)	0.81 (0.77)	0.76 (0.71)	**0.75 (0.75)**	**0.89 (0.89)**	**0.91 (0.91)**	**0.86 (0.86)**
U-MtE + balanced (ours)	ROI + A	0.69 (0.69)	0.63 (0.63)	**0.79 (0.79)**	**0.79 (0.79)**	**0.75 (0.75)**	0.73 (0.73)	0.84 (0.84)	0.84 (0.84)	0.80 (0.80)

Erasure. The insertion eraser removes the span of the differences:

$$\Sigma_\Delta = \tfrac{1}{n}\textstyle\sum_i \Delta_i \Delta_i^\top, \qquad U_k = \text{top-}k \text{ eigvecs}(\Sigma_\Delta), \qquad P(z) = z - U_k U_k^\top (z - \mu), \tag{1}$$

with k chosen *without labels* as the smallest rank whose held-out paired-difference energy $\mathbb{E}\|P\Delta\|^2/\mathbb{E}\|\Delta\|^2$ falls below 0.10 (fixed before any experiment).

Mask-then-erase (U-MtE). Masking and erasure have complementary reach: masking removes any artifact outside the ROI but none inside; erasure acts on the representation regardless of location. U-MtE therefore computes the pairs *on masked images* ($\Delta_i = f(m(x_i^+)) - f(m(x_i))$) so that the erased subspace targets exactly what masking leaves behind, and fits the head on $P(f(m(x)))$. When image-level artifact labels exist, the head is group-balanced (U-MtE$_{\text{bal}}$).

Disease protection. If the artifact resembles the pathology (intestinal debris versus erosion fibrin), U_k can contain disease evidence. Let W be an orthonormal basis of logistic label directions fitted on artifact-free training images; we erase $U_k' = \text{orth}((I - WW^\top)U_k)$ instead, which guarantees $w^\top P(z) = w^\top z$ for every $w \in \text{span}(W)$ (cf. task-preserving erasure, 6).

Scope. U-MtE is a representation-level intervention: it acts on the features of a frozen encoder, the dominant way in which foundation models are deployed on small medical datasets, and needs only the encoder, the ROI mask and the generator. An end-to-end analogue is evaluated as a secondary check (S8.3).

Tool. The released package exposes insertion, erasure and a patch-token artifact probe for any frozen backbone (ViTs and CNNs) through a two-call API.

S8.2 Results on the traps

U-MtE repairs what masking leaves behind. Without any artifact example, artifact mask or artifact label—it uses only the ROI mask that masking already requires—U-MtE improves reversed AUROC over masking

in the in-ROI regime of every modality: thyroid calipers +0.224 [+0.195, +0.253] (DINOv2) and +0.246 [+0.224, +0.269] (MedSigLIP); dermoscopic hair +0.040 [+0.024, +0.056]; synthetic capsule debris +0.121 [+0.078, +0.170]; synthetic calipers +0.278 [+0.214, +0.344]; synthetic chest tubes +0.134 [+0.113, +0.156] (DINOv2 / RAD-DINO). With MedSigLIP the unbalanced method *loses* to masking in both controlled sweeps (debris −0.190 [−0.265, −0.120], calipers −0.076 [−0.120, −0.034]) although it wins on real thyroid calipers (+0.246); the balanced variant beats masking at full overlap in all four controlled sweeps (+0.211 to +0.470) and has no significant location interaction in three of them (with MedSigLIP on capsule it gains *more* at full overlap: interaction −0.165 [−0.300, −0.025]). Disease protection turns the MedSigLIP debris failure into a gain (+0.114 [+0.022, +0.238] over masking at full overlap) and the caliper one into a non-significant difference (−0.041 [−0.095, +0.002]).

Sensitivity to the erasure rank. In a pre-registered ablation (Trap A, DINOv2), the thyroid gain over masking is +0.18 to +0.22 for every setting that erases at least 16 directions (held-out energy 0.5–0.99, or fixed $k \in \{16, 64\}$), but only +0.06 to +0.07 with one to four directions: the shift caused by overlays is multi-dimensional. The pre-specified 90% rule reached the rank cap ($k = 64$) in all three cohorts. The protected variant never harms capsule by more than 0.010 and helps ovary at every rank (+0.007 to +0.043), whereas unprotected erasure harms capsule more the more directions it removes, as the overlap term ρ of the linear-Gaussian account (main paper) predicts.

Erasure, not augmentation. Using the same generic overlays as training augmentation (overlaid copies of a random half of the training images, in the masked view) gains at most +0.03 over masking; erasing their subspace instead gains +0.215 [+0.189, +0.244] (thyroid, DINOv2), +0.225 [+0.202, +0.247] (thyroid, MedSigLIP) and +0.035 [+0.017, +0.055] (hair) more than augmentation. Under the stricter near-duplicate grouping (main paper, stricter leakage grouping) the thyroid advantage is unchanged (+0.192) but the small ovary advantage (+0.035) loses significance, so we regard erase > augment as established for thyroid and hair only.

When the artifact resembles the disease. On real capsule debris U-MtE is worse than masking (−0.126 [−0.146, −0.107]): erosions carry yellow-white fibrin that resembles debris, and a subspace estimated from real-debris templates is worse still (−0.392 [−0.425, −0.360]). Disease protection removes the failure (−0.006 [−0.018, +0.006] generic; template variant +0.003 [−0.014, +0.021]) at no cost where erasure already works (protected minus unprotected: thyroid +0.059 [+0.019, +0.096], hair +0.001 [−0.016, +0.017]).

With image-level artifact labels. Adding group balancing, U-MtE$_{\text{bal}}$ is the most robust model in thyroid ultrasound by the minimum of correlated and reversed AUROC (0.788 DINOv2, 0.792 MedSigLIP; DFR 0.659 and 0.719); in real capsule endoscopy masking plus last-layer retraining is higher (0.885 versus 0.857);

on dermoscopic hair DFR alone is best (0.741) because most of the hair shortcut is not in the hair pixels (removing every hair patch with the oracle mask gains only +0.015). In both controlled sweeps the benefit of U-MtE$_{\text{bal}}$ no longer depends on location (interaction +0.021 $[-0.019, +0.061]$ capsule; −0.053 $[-0.108, +0.003]$ thyroid), whereas that of masking does (+0.228, +0.433). The capsule findings replicate with MedSigLIP (crossover +0.335 $[+0.286, +0.383]$; masking plus DFR 0.901). All thyroid and capsule findings replicate with a CNN (ConvNeXt-Base): crossovers +0.275 and +0.320; U-MtE − mask +0.135 $[+0.102, +0.168]$ and erase − augment +0.128 $[+0.093, +0.161]$ on thyroid.

Label-free and erasure baselines. Among methods that need no artifact labels, U-MtE beats JTT [8] by +0.206 $[+0.171, +0.241]$ (thyroid), +0.125 $[+0.063, +0.189]$ (ovary) and +0.094 $[+0.077, +0.111]$ (chest drains, RAD-DINO; not with DINOv2, −0.005 $[-0.021, +0.011]$); JTT on the unmasked view is best on hair. Task-preserving concept removal (SPLINCE; 6), fitted with artifact labels in the whitened, minimal-distortion form of the original method, removes the artifact covariance at a large cost in disease signal (clean AUROC 0.56–0.78 versus 0.69–0.87 for masking) and still flips the shortcut on capsule endoscopy (reversed 0.868, correlated 0.683): when artifact and label are strongly associated in training, removing one removes much of the other. By min(reversed, correlated) U-MtE$_{\text{bal}}$ exceeds SPLINCE on thyroid, capsule and hair (0.788 vs 0.596; 0.857 vs 0.683; 0.704 vs 0.564); on chest drains SPLINCE is higher (0.609, archived run, vs 0.461) but below DFR (0.666).

Text-prompted erasure. With vision–language backbones an artifact direction can be obtained from text alone ("a thyroid ultrasound image with measurement calipers" minus "a thyroid ultrasound image"; biased-prompt construction, 2). Erasing that subspace after masking is a near no-op in every cohort (+0.001 to +0.008 over masking; MedSigLIP thyroid, ovary, capsule and DermLIP hair; archived runs, point estimates), whereas U-MtE with the same MedSigLIP features gains +0.246 $[+0.224, +0.269]$ (thyroid) and +0.210 $[+0.167, +0.253]$ (ovary). The direction along which a described artifact moves the *text* embedding is not the direction along which the real artifact moves the *image* embedding; synthetic insertions measure the latter directly.

A single procedure: validation-driven selection. Selecting, per fold, the candidate with the best group-aware validation score (image-level artifact labels needed on validation images only) beats masking in all five cohorts (+0.17 to +0.40), is within 0.02 of the best candidate in each, and beats DFR in three (thyroid +0.057, capsule +0.047, ovary +0.010; archived runs, point estimates); it chooses U-MtE$_{\text{bal}}$ where the artifact is an overlay and plain balancing on hair and drains, where DFR remains better. A split-validation variant that admits DFR as a candidate is reported in the supplement.

Modern inpainting. Inpainting the detected caliper pixels with LaMa [9] and then masking is a strong baseline when artifact masks are available (over masking: thyroid +0.224 $[+0.206, +0.242]$, ovary

+0.128 [+0.084, +0.169]). U-MtE$_{\text{protect}}$, which needs no artifact mask, exceeds it on thyroid (+0.060 [+0.013, +0.105]) and is below it on ovary (−0.085 [−0.120, −0.050]) and on dermoscopic hair, where published hair masks are available (−0.086; archived run, point estimate); with image-level artifact labels, U-MtE$_{\text{bal}}$ exceeds it in all three (+0.165 [+0.146, +0.185], +0.061 [+0.026, +0.095], +0.123, archived). The thyroid comparison changed when the run was regenerated with per-image predictions (first run: −0.009): the protected arm reproduced about 0.06 higher, with unchanged code for that arm, and now agrees with the protected arm of the main regenerated run (`results/bootstrap_correction/archived_old_vs_new.md`).

S8.3 End-to-end fine-tuning and erase-while-fine-tuning

Fine-tuning ResNet-50 end-to-end (five folds) reproduces the frozen-backbone findings. Capsule: masking harms in-ROI (−0.279 [−0.377, −0.185]) and helps out-of-ROI (+0.301), crossover +0.606 [+0.495, +0.716]. Thyroid: the crossover points the same way (+0.063 [−0.022, +0.144], five folds), and an end-to-end analogue of U-MtE—an invariance penalty between masked images with and without a generic overlay—improves on masking by +0.139 [+0.096, +0.184] without shortcut flipping (reversed 0.679, correlated 0.703), matching the best label-using arm. It fails on capsule (−0.021), where the artifact resembles the pathology. Ovary (1,202 images) is underpowered: U-MtE end-to-end has the best reversed, correlated and clean AUROC, with CIs overlapping masking. A fine-tuned ViT-S reproduces the crossover on thyroid (+0.122 [+0.039, +0.211]), but the end-to-end U-MtE analogue does not improve on masking with this architecture (−0.000 [−0.026, +0.026]). Applying U-MtE post hoc to the penultimate features of the masked, fine-tuned ResNet-50 does not help either (pre-registered; −0.000 [−0.001, +0.001] over the same features without erasure): after task training the shortcut is encoded along task-specific directions that generic overlays do not move. For fully fine-tuned networks the subspace must be estimated and removed *during* training. *Erase-while-fine-tuning* (pre-registered; no artifact annotation) projects a running estimate of the overlay-shift subspace of the network's own penultimate features out before the head at every step and at test time, and adds feature invariance and prediction consistency between masked images with and without a generic overlay. On thyroid it improves on masking by +0.195 [+0.153, +0.237] with ResNet-50 and +0.119 [+0.036, +0.198] with ViT-S (reversed AUROC 0.693 and 0.608 vs 0.536 and 0.465), whereas each component alone works for only one architecture (projection + invariance: ResNet-50 +0.186 [+0.144, +0.230], ViT-S +0.032 [−0.093, +0.144]; prediction consistency: ResNet-50 +0.042 [+0.011, +0.074], ViT-S +0.061 [+0.036, +0.084]; with the corrected bootstrap consistency alone is now significant for both, but smaller than the combination). With ResNet-50 the correlated AUROC falls slightly below the reversed one (0.661 vs 0.693), a mild over-correction. On ovary (ResNet-50) it is positive but not significant (+0.045 [−0.036, +0.130]; every fine-tuning contrast on this 1,202-image cohort is underpowered), and on capsule, where debris resembles erosion fibrin, it fails (−0.026 [−0.056, +0.006]), as does every annotation-free method.

Table 3: Masking versus ERM on the unaltered, patient-disjoint official thyroid test split (614 images, 236 malignant) at five operating points fixed on validation data (mean over five seeds; Δ = mask − ERM with 95% crossed seed×image bootstrap CI, thresholds re-estimated in every replicate). *Conflicting*: the 78 malignant nodules with an in-ROI caliper (the caliper marks benign nodules in this dataset). The subgroup was defined after the first unaltered-data analysis; the operating-point test in it was registered before these numbers were computed.

Threshold on validation	Sens. ERM → mask	Δ sensitivity	Sens. conflicting ($n = 78$)	Δ sensitivity, conflicting	Δ specificity
max. balanced accuracy	0.699 → 0.424	−0.275 [−0.435, −0.138]	0.703 → 0.223	−0.479 [−0.640, −0.313]	+0.202 [+0.115, +0.330]
specificity ≥ 0.80	0.557 → 0.355	−0.202 [−0.300, −0.100]	0.549 → 0.177	−0.372 [−0.499, −0.236]	+0.117 [+0.059, +0.172]
specificity ≥ 0.90	0.371 → 0.219	−0.152 [−0.253, −0.043]	0.336 → 0.100	−0.236 [−0.383, −0.089]	+0.075 [+0.029, +0.122]
sensitivity ≥ 0.80	0.753 → 0.380	−0.373 [−0.469, −0.270]	0.744 → 0.197	−0.546 [−0.671, −0.421]	+0.278 [+0.203, +0.359]
sensitivity ≥ 0.90	0.869 → 0.588	−0.281 [−0.391, −0.178]	0.862 → 0.400	−0.462 [−0.614, −0.310]	+0.339 [+0.236, +0.425]

S9. Operating points on every unaltered test set

Masking minus ERM at the five pre-registered validation-fixed operating points (95% crossed seed×image bootstrap CIs; thresholds re-estimated in every replicate). The first table gives the official thyroid split in full, including the conflicting subgroup; the second every unaltered test set. *Conflicting*: positive cases whose in-ROI artifact status points to the negative class. Generated by `scripts/analysis/operating_points.py`.

Test set	Threshold (validation)	Δ sensitivity	Δ sensitivity, conflicting	Δ specificity
Thyroid	maxBA	−0.275 [−0.435, −0.138]	−0.479 [−0.640, −0.313]	+0.202 [+0.115, +0.330]
	spec0.80	−0.202 [−0.300, −0.100]	−0.372 [−0.499, −0.236]	+0.117 [+0.059, +0.172]
	spec0.90	−0.152 [−0.253, −0.043]	−0.236 [−0.383, −0.089]	+0.075 [+0.029, +0.122]
	sens0.80	−0.373 [−0.469, −0.270]	−0.546 [−0.671, −0.421]	+0.278 [+0.203, +0.359]
	sens0.90	−0.281 [−0.391, −0.178]	−0.462 [−0.614, −0.310]	+0.339 [+0.236, +0.425]
ISIC, BCN held out	maxBA	+0.000 [−0.044, +0.055]	+0.001 [−0.045, +0.059]	−0.078 [−0.174, +0.013]
	spec0.80	+0.017 [−0.003, +0.036]	+0.019 [−0.004, +0.042]	−0.110 [−0.135, −0.079]
	spec0.90	+0.042 [+0.012, +0.071]	+0.042 [+0.006, +0.076]	−0.104 [−0.135, −0.074]
	sens0.80	+0.002 [−0.029, +0.035]	+0.003 [−0.030, +0.038]	−0.082 [−0.138, −0.020]
	sens0.90	+0.001 [−0.029, +0.030]	−0.001 [−0.036, +0.034]	−0.065 [−0.134, +0.016]
ISIC, HAM held out	maxBA	−0.006 [−0.117, +0.150]	+0.002 [−0.110, +0.165]	+0.042 [−0.070, +0.124]
	spec0.80	−0.062 [−0.110, −0.015]	−0.055 [−0.104, −0.007]	+0.092 [+0.064, +0.121]
	spec0.90	−0.057 [−0.115, −0.005]	−0.053 [−0.113, +0.004]	+0.041 [+0.020, +0.066]
	sens0.80	−0.004 [−0.055, +0.044]	+0.009 [−0.049, +0.062]	+0.061 [+0.017, +0.114]
	sens0.90	−0.021 [−0.060, +0.017]	−0.013 [−0.056, +0.031]	+0.096 [+0.041, +0.160]
ISIC, MSK held out	maxBA	−0.013 [−0.108, +0.080]	−0.033 [−0.129, +0.062]	+0.032 [−0.033, +0.104]
	spec0.80	−0.074 [−0.127, −0.026]	−0.094 [−0.149, −0.044]	+0.083 [+0.064, +0.110]
	spec0.90	−0.091 [−0.145, −0.043]	−0.110 [−0.167, −0.058]	+0.058 [+0.042, +0.075]
	sens0.80	−0.050 [−0.105, +0.000]	−0.070 [−0.129, −0.017]	+0.068 [+0.037, +0.105]
	sens0.90	−0.023 [−0.080, +0.029]	−0.037 [−0.096, +0.017]	+0.073 [+0.024, +0.124]
Capsule	maxBA	+0.086 [+0.017, +0.169]	−0.074 [−0.167, +0.048]	+0.089 [+0.031, +0.152]
	spec0.80	+0.140 [+0.082, +0.222]	+0.022 [−0.041, +0.123]	+0.041 [−0.029, +0.107]
	spec0.90	+0.245 [+0.159, +0.316]	+0.092 [−0.039, +0.214]	−0.000 [−0.039, +0.057]
	sens0.80	+0.030 [−0.014, +0.080]	−0.096 [−0.201, −0.006]	+0.092 [+0.045, +0.148]
	sens0.90	+0.022 [−0.026, +0.072]	−0.104 [−0.204, −0.026]	+0.169 [+0.092, +0.275]

S10. Absolute performance and published results

Only the thyroid cohort has a published benchmark on the same split and labels (official patient-disjoint TNCD split, 614 test images). The ovarian (benign cystic versus solid-component tumours) and capsule (erosion versus polyp-like) binary tasks and the ISIC 2019 leave-one-hospital-out melanoma task were defined for this study and have no published counterpart; their clean AUROC is reported with every result table. Frozen encoders with a linear head are below fine-tuned CNNs; the location law replicates with fine-tuned networks (main text).

Model	Source	Test AUROC
ResNet-18, cross-entropy (fine-tuned)	Gong et al. [5]	0.773
DenseNet-121, cross-entropy (fine-tuned)	Gong et al. [5]	0.780
ConvNeXt-T, cross-entropy (fine-tuned)	Gong et al. [5]	0.784
ResNet-18, adaptive curriculum (fine-tuned)	Gong et al. [5]	0.799
DINOv2 ViT-B/14 frozen + linear head, ERM (rebuilt run)	this work	0.735

S11. Covariate balance of the traps and the real-artifact transplant

Full matched analysis (pre-registered calliper 0.2 SD, and 0.05 SD as a sensitivity analysis).

Cohort	Calliper (SD)	In-ROI / out-of-ROI images	Kept per trap (neg./pos.)	$\lvert SMD \rvert_{max}$ before	after (pooled)	after (within label)	Crossover unmatched	Crossover matched
ISIC hair	0.20	5,438 / 8,329	1,829 (1431/398)	2.13	0.11	0.16	+0.147 [+0.111, +0.185]	+0.154 [+0.123, +0.186]
	0.05	5,438 / 8,329	1,777 (1394/383)	2.13	0.13	0.18	+0.147 [+0.111, +0.185]	+0.124 [+0.086, +0.162]
Thyroid	0.20	1,150 / 326	316 (223/93)	0.53	0.04	0.15	+0.239 [+0.196, +0.280]	+0.238 [+0.186, +0.290]
	0.05	1,150 / 326	299 (212/87)	0.53	0.06	0.19	+0.239 [+0.196, +0.280]	+0.218 [+0.165, +0.270]
Ovary	0.20	617 / 194	179 (86/93)	0.39	0.08	0.14	+0.170 [+0.087, +0.252]	+0.163 [+0.050, +0.273]
	0.05	617 / 194	165 (82/83)	0.39	0.08	0.18	+0.170 [+0.087, +0.252]	+0.152 [+0.041, +0.265]
Capsule	0.20	816 / 1,240	71 (19/52)†	2.57	0.23	0.49	+0.377 [+0.327, +0.426]	+0.230 [+0.036, +0.434]
	0.05	816 / 1,240	56 (11/45)†	2.57	0.23	0.67	+0.377 [+0.327, +0.426]	+0.328 [+0.087, +0.573]

†Infeasible (fewer than 30 pairs in one label). Crossed seed×image CIs. *Doubly robust analysis (pre-registered).* Per-image placement values of the masked minus the ERM score, regressed within seed on the trap indicator, the label and every matching covariate on the matched samples, with a crossed bootstrap (2,000 replicates): hair +0.156 [+0.127, +0.187], thyroid +0.238 [+0.184, +0.291], ovary +0.165 [+0.056, +0.273] (Holm $p = 0.002$ each). Within a label some covariates keep standardised differences of 0.13–0.19 after matching, so residual confounding by them cannot be excluded; the transplant does not depend on it. *Paste-edge control.* Neutral tissue from the donor image (or, where no artifact-free window exists, from an artifact-free image of the cohort) pasted through the same masks, transforms, positions and blending as the artifact.

Table 4: Covariate balance of the location traps: standardised mean differences (SMD) between the artifact-bearing images of Trap A (in-ROI) and Trap B (out-of-ROI), before and after propensity matching (categorical covariates: largest level). Positive: larger in Trap A.

Cohort	Covariate	SMD before	SMD after
ISIC hair	lesion area	+2.13	+0.03
	hair amount	-0.16	+0.11
	age	+0.14	-0.01
	age missing	-0.04	-0.05
	sex (missing)	-0.03	-0.04
	site (torso)	+0.19	-0.01
	benign dx (MEL)	+0.26	+0.00
	source (MSK)	-0.54	+0.00
Thyroid	roi area	+0.40	-0.01
	caliper px log	+0.53	+0.04
	width	-0.14	+0.04
	height	-0.12	+0.02
	mean grey	+0.18	-0.01
	roi centroid row	+0.06	+0.01
Ovary	roi area	+0.32	-0.00
	caliper px log	+0.39	+0.03
	width	-0.30	+0.04
	height	+0.05	+0.00
	mean grey	+0.11	+0.02
	subtype (2.0)	-0.16	-0.01
Capsule	roi area	+2.57	+0.01
	contamination	-0.33	-0.12
	mean grey	+0.13	+0.11
	frame block (0.0)	-0.37	-0.23

Cohort	Artifact	Neutral paste	Neutral / artifact	Artifact − neutral	p_{Holm}	Attribu
ISIC hair	+0.253 [+0.214, +0.293]	+0.204 [+0.164, +0.244]	0.81	+0.049 [+0.026, +0.072]	0.0004	
Thyroid	+0.277 [+0.256, +0.298]	+0.145 [+0.130, +0.161]	0.52	+0.132 [+0.116, +0.148]	0.0004	
Ovary	+0.074 [+0.044, +0.104]	+0.052 [+0.021, +0.084]	0.71	+0.022 [−0.002, +0.048]	0.1372	
Capsule	+0.497 [+0.427, +0.567]	+0.459 [+0.383, +0.535]	0.92	+0.037 [−0.017, +0.093]	0.1792	

Pre-registered rule: attributed to the artifact only if artifact − neutral > 0 (Holm) and the neutral interaction is below one quarter of the artifact's.

Transplant design. Recipients: artifact-free images of each cohort. Donors: artifact-bearing images of the same cohort; an artifact instance is the largest connected component of its mask after a grouping dilation (calipers 9 px, hair and debris 5 px), with at least 30 pixels and a bounding-box side of at most 259 px. For every recipient, donors are tried in a stable random order (up to 50 donors × 8 dihedral transforms) until one fits both inside (overlap ≥ 0.95) and outside (overlap 0) the ROI within the field of view; the instance is alpha-composited with a Gaussian-blurred mask. En-

Table 5: Real-artifact transplant: reversed / correlated / clean AUROC of ERM and ROI masking with the same real artifact inside or outside the ROI of the same images, and the change from masking (95% CI). $|\Delta p|$: same-head counterfactual sensitivity to inserting the artifact.

Cohort	Location	ERM rev / corr / clean	Mask rev / corr / clean	Mask − ERM (rev)	$\lvert\Delta p\rvert$ ERM / mask
ISIC hair	outside ROI	0.539 / 0.869 / 0.759	0.716 / 0.716 / 0.716	$+0.178$ $[+0.128, +0.229]$	0.230 / 0.000
	inside ROI	0.454 / 0.904 / 0.776	0.379 / 0.842 / 0.672	-0.075 $[-0.115, -0.036]$	0.340 / 0.297
Thyroid	outside ROI	0.497 / 0.857 / 0.708	0.846 / 0.846 / 0.846	$+0.349$ $[+0.321, +0.378]$	0.201 / 0.000
	inside ROI	0.487 / 0.851 / 0.708	0.559 / 0.906 / 0.801	$+0.072$ $[+0.044, +0.100]$	0.206 / 0.271
Ovary	outside ROI	0.686 / 0.759 / 0.725	0.750 / 0.750 / 0.750	$+0.064$ $[+0.002, +0.127]$	0.023 / 0.000
	inside ROI	0.657 / 0.771 / 0.723	0.647 / 0.825 / 0.755	-0.010 $[-0.075, +0.057]$	0.037 / 0.116
Capsule	outside ROI	0.592 / 0.905 / 0.795	0.914 / 0.914 / 0.914	$+0.322$ $[+0.264, +0.383]$	0.193 / 0.000
	inside ROI	0.508 / 0.906 / 0.787	0.333 / 0.956 / 0.871	-0.175 $[-0.241, -0.108]$	0.238 / 0.551

vironments as in the controlled sweeps (0.9/0.1 training and correlated test, 0.1/0.9 reversed test, clean test without artifact), 5-fold grouped cross-validation, five presence seeds (bootstrap clusters). Recipients without a feasible donor (mostly small capsule lesion boxes) are excluded; counts by label are in `results/review2/transplant/*/feasibility_by_label.csv`.

S12. CLAIM 2024 checklist

Checklist for Artificial Intelligence in Medical Imaging, 2024 update [10]. M = main paper section, S = supplement section.

No.	Item	Answer	Where / comment
Title / abstract			
1	Identification as a study of AI methodology, specifying the category of technology used (eg, deep learning)	Yes	Title, abstract: deep-learning classifiers on frozen foundation-model encoders and fine-tuned CNN/ViT.
2	Summary of study design, methods, results, and conclusions	Yes	Abstract.
Introduction			
3	Scientific and/or clinical background, including the intended use and role of the AI approach	Yes	M Introduction, Related work.
4	Study aims, objectives, and hypotheses	Yes	M Introduction; hypotheses pre-registered (S1).
Methods — study design			
5	Prospective or retrospective study	Yes	Retrospective analysis of public datasets (M Data).
6	Study goal	Yes	Methodological study of shortcut mitigation (not a diagnostic-accuracy study); M Introduction, Setup.
Methods — data			
7	Data sources	Yes	M Data (ISIC 2018/2019, HAM10000 masks, TN3K/TNCD, MMOTU, SEE-AI, NIH ChestX-ray14); licences in `docs/DATA.md`.

No.	Item	Answer	Where / comment
8	Inclusion and exclusion criteria	Yes	M Data (trap cells, artifact-free groups, QC); transplant feasibility (S11).
9	Data preprocessing	Yes	M Setup (518 px rendering, ROI masking, feature caching).
10	Selection of data subsets	Yes	Traps, matched traps and transplant cohorts (M Results, S11).
11	De-identification methods	NA	Public datasets released de-identified by their providers.
12	How missing data were handled	Yes	Missing age imputed with an indicator; images without a lesion mask excluded by QC (152 ISIC images).
13	Image acquisition protocol	Partial	As documented by the dataset providers (cited); no acquisition parameters are released for TN3K, MMOTU or SEE-AI.
Methods — reference standard			
14	Definition of method(s) used to obtain reference standard	Yes	Diagnoses as released by the dataset providers (TNCD: cytological biopsy; ISIC: histopathology, follow-up or expert consensus; MMOTU tumour categories; SEE-AI expert lesion annotation) (M Data).
15	Rationale for choosing the reference standard	Yes	Dataset reference standards; ROI masks from manual annotation or a U-Net (M Data).
16	Source of reference standard annotations	Yes	Dataset annotators; artifact labels from a rule-based detector, a patch probe and published masks; full human/automatic provenance in S4.
17	Annotation of test set	Yes	Same procedures as for training images.
18	Measures of inter- and intrarater variability of features described by the annotators	No	Artifact labels were audited by the analysis team only; a blinded two-rater expert audit kit with Cohen's κ is prepared (S4) but not yet carried out (M Limitations).
Methods — data partitions			
19	How data were assigned to partitions	Yes	Five seeds of five-fold group-safe cross-validation; official TNCD split; leave-one-hospital-out (M Setup, Consequences on unaltered data).
20	Level at which partitions are disjoint	Yes	Lesion identifiers (ISIC), patient level for the official TNCD split and chest radiographs, otherwise perceptual-hash and frame-block groups; embedding-group sensitivity (M Robustness).
Methods — testing data			
21	Intended sample size	No	No a-priori sample-size calculation; all eligible images used; underpowered contrasts are identified (ovary).
Methods — model			
22	Detailed description of model	Yes	Frozen DINOv2 ViT-S/B/L, MedSigLIP, DermLIP, RAD-DINO, ConvNeXt with logistic heads; fine-tuned ResNet-50 and ViT-S (M Setup, Replication; S8).
23	Software libraries, frameworks, and packages	Yes	`requirements.txt` (PyTorch 2.11, timm, scikit-learn, open_clip) (S13).
24	Initialization of model parameters	Yes	Published pre-trained checkpoints; ImageNet initialisation for fine-tuning.
Methods — training			
25	Details of training approach	Yes	Regularisation and decision threshold on clean validation data; fine-tuning recipe (S8).
26	Method of selecting the final model	Yes	Clean-validation AUROC; no test-set selection.
27	Ensembling techniques	NA	No ensembles.
Methods — evaluation			

No.	Item	Answer	Where / comment
28	Metrics of model performance	Yes	Reversed, correlated and clean AUROC; location crossover; AUPRC, Brier, ECE, sensitivity/specificity (M Setup, Consequences on unaltered data).
29	Statistical measures of significance and uncertainty	Yes	Crossed seed×image bootstrap for every model-comparison interval, from regenerated runs (S15; old-versus-new comparison in the repository); image bootstrap for label agreement; point estimates only for the archived runs without per-image predictions; Holm per family (M Setup; S1).
30	Robustness or sensitivity analysis	Yes	Label definitions, leakage grouping, covariate matching, rebuild on new hardware (M Robustness).
31	Methods for explainability or interpretability	Yes	Same-head counterfactual sensitivity to inserting the artifact (M Mechanism).
32	Evaluation on internal data	Yes	All traps and sweeps.
33	Testing on external data	Partial	Dermoscopy: new patients from ISIC 2020 under a pre-registered gate (patient-disjoint folds) and ISIC leave-one-hospital-out; no external cohort for ultrasound or capsule endoscopy (M Limitations; S3).
34	Clinical trial registration	NA	Retrospective; analysis pre-registrations in the repository (S1).
Results			
35	Numbers of patients or examinations included and excluded	Yes	Image counts per cohort, trap cell and transplant feasibility; patients where identifiers exist (M Data; S11).
36	Demographic and clinical characteristics of cases in each partition	Partial	Age, sex, body site and diagnosis for ISIC; lesion size and acquisition covariates for all cohorts (S11); demographics are not released for TN3K, MMOTU or SEE-AI.
37	Performance metrics and measures of statistical uncertainty	Yes	All result tables give 95% CIs.
38	Estimates of diagnostic performance and their precision	Yes	Sensitivity and specificity at five pre-registered operating points with CIs (M Consequences on unaltered data; S9).
39	Failure analysis of incorrectly classified cases	Yes	Shortcut-conflicting cases; DermLIP without a hair gap; chest radiography boundary case (M Results).
Discussion			
40	Study limitations	Yes	M Discussion and limitations.
41	Implications for practice, including intended use and/or clinical role	Yes	M Discussion; decision guide (M What carries the shortcut decides the fix).
Other information			
42	Provide a reference to the full study protocol or to additional technical details	Yes	Pre-registration documents and registry (S1); `report/full/full_report.pdf`.
43	Statement about the availability of software, trained model, and/or data	Yes	Code, pre-registrations and result files are public; all datasets are public (M Conclusion).
44	Sources of funding and other support; role of funders	—	To be completed by the authors.

S13. Reproducibility

All numbers in the main paper and this supplement are produced by scripts in the repository (`scripts/`); tables are generated, never typed, and every confidence interval in the paper is traced automatically to a result file (`scripts/verify/audit_paper_numbers.py`). A complete rebuild on different hardware reproduces

Table 7: Synthetic ruler, ISIC 2018, DINOv2 ViT-B/14 @518: AUROC and reversed-test Δ vs ERM [95% CI]

Method	Needs	clean@0	rev@0	clean@50	rev@50	clean@100	rev@100	Δrev@0	Δrev@50	Δrev@100
ERM	—	0.797	0.665	0.793	0.573	0.792	0.515			
ROI masking	ROI mask (train+test)	0.849	0.849	0.829	0.475	0.825	0.386	+0.184 [+0.117, +0.250]	-0.097 [-0.164, -0.030]	-0.129 [-0.195, -0.064]
Inpainting (oracle)	artifact pixels (train+test)	0.810	0.806	0.809	0.783	0.807	0.758	+0.141 [+0.109, +0.175]	+0.211 [+0.175, +0.249]	+0.243 [+0.205, +0.284]
Group-balanced	image-level A (train)	0.814	0.802	0.815	0.791	0.806	0.796	+0.137 [+0.099, +0.179]	+0.219 [+0.177, +0.265]	+0.281 [+0.236, +0.329]
DFR	image-level A (val)	0.786	0.789	0.787	0.794	0.789	0.794	+0.124 [+0.075, +0.179]	+0.221 [+0.166, +0.281]	+0.279 [+0.220, +0.339]
LEACE (paired, oracle renders)	paired renders (train)	0.829	0.832	0.810	0.810	0.809	0.809	+0.167 [+0.126, +0.211]	+0.237 [+0.197, +0.280]	+0.294 [+0.249, +0.339]
Repair consistency	artifact pixels (train)	0.828	0.709	0.822	0.616	0.814	0.557	+0.044 [-0.002, +0.091]	+0.044 [-0.007, +0.092]	+0.042 [-0.009, +0.092]
Insert-then-erase (artifact templates)	artifact templates	0.849	0.844	0.850	0.833	0.847	0.818	+0.179 [+0.136, +0.225]	+0.260 [+0.212, +0.311]	+0.303 [+0.250, +0.357]
Insert-then-erase + balanced (artifact templates)	artifact templates + A (train)	0.841	0.842	0.843	0.840	0.844	0.842	+0.177 [+0.135, +0.221]	+0.267 [+0.219, +0.316]	+0.327 [+0.276, +0.380]
Insert-then-erase rank-1 (ablation)	artifact templates	0.811	0.799	0.809	0.770	0.806	0.730	+0.134 [+0.104, +0.166]	+0.197 [+0.161, +0.235]	+0.215 [+0.178, +0.254]
Insertion augmentation	artifact templates	0.814	0.726	0.812	0.653	0.810	0.603	+0.061 [+0.045, +0.079]	+0.081 [+0.064, +0.099]	+0.088 [+0.072, +0.105]

every primary crossover within 0.01 (S18). Compute: one 24 GB GPU.

S14. Controlled sweeps, mechanism and theory in detail

Controlled dermoscopy. With a synthetic ruler whose overlap with the lesion is swept from 0 to 100% (2,437 ISIC 2018 images, DINOv2 ViT-B/14 at 518 px, three seeds), ROI masking improves reversed-environment AUROC over ERM by $+0.184$ $[+0.117, +0.250]$ when the ruler lies outside the lesion and *lowers* it by -0.097 $[-0.164, -0.030]$, -0.111 $[-0.181, -0.043]$ and -0.129 $[-0.195, -0.064]$ at 50, 75 and 100% overlap (location interaction $+0.313$ $[+0.258, +0.370]$; Table 7); at 25% overlap the change is not significant (-0.059 $[-0.127, +0.008]$). The sign flip is reproduced at 224 px and under randomised ruler colour, opacity, ticks and scale. At 0% overlap the masked model's correlated and reversed AUROC coincide (0.849/0.849); at 100% its correlated AUROC exceeds ERM's (0.961 vs 0.937): it uses the ruler *more*. Synthetic calipers on thyroid nodules, synthetic debris on capsule lesions and synthetic calipers on ovarian tumours give the same interaction (thyroid $+0.433$ $[+0.362, +0.507]$, capsule $+0.228$ $[+0.144, +0.324]$, ovary $+0.238$ $[+0.144, +0.340]$; main-text figure of the controlled sweeps); whether the lost benefit becomes harm depends on the setting (ovary at full overlap -0.197 $[-0.357, -0.035]$; thyroid -0.030 $[-0.105, +0.042]$).

Mechanism in detail. Three explanations were tested separately. *Occlusion*: with the ruler present but uncorrelated with the label (50/50), masking changes AUROC by $+0.036$, $+0.041$ and $+0.039$ at 0/50/100% overlap, none of them significantly; covering lesion pixels costs nothing measurable, so the harm requires the correlation. *Lost context*: dilating the mask at 50% overlap *increases* harm ($-0.097 \rightarrow -0.138 \rightarrow -0.153$ for margins 0/10/25 px; neighbouring intervals overlap) as the margin restores ruler pixels (retention $0.50 \rightarrow 0.73 \rightarrow 0.95$); at 100% overlap, where retention is already 1, harm is flat (-0.129 to -0.144) although the ruler's share of visible pixels falls threefold. *Amplified reliance*: masking doubles the same-head counterfactual sensitivity to the ruler ($|\Delta p|$ $0.235 \rightarrow 0.491$ at 100%). Harm is largest for small lesions (-0.364 $[-0.509, -0.217]$ at 100%), where the ruler is a larger share of what survives the mask; for large lesions the effect at 100% is not significant ($+0.063$ $[-0.015, +0.144]$), and at 50% overlap only the small-lesion stratum is significant, so lesion size orders the harm without defining a threshold. The main-text mechanism figure shows the same retention mechanism in all five controlled sweeps: with the head held

fixed, inserting the artifact changes the masked model's prediction more the deeper the artifact lies in the ROI. At full overlap this sensitivity exceeds that of ERM for dermoscopic rulers (0.491 vs 0.235; paired difference $+0.255$ $[+0.221, +0.287]$), ovarian calipers (0.218 vs 0.032; $+0.188$ $[+0.133, +0.250]$) and capsule debris (0.245 vs 0.057; $+0.186$ $[+0.130, +0.260]$) and matches it for thyroid calipers (0.393 vs 0.369; $+0.025$ $[-0.016, +0.065]$). In chest radiography ERM ignores a tube outside the lungs ($|\Delta p| = 0.005$ at $r = 0$), which is why masking has nothing to remove there. Erasing a generic-overlay subspace after masking and group balancing both lower this sensitivity in every sweep (S8).

What the real-hair shortcut is made of. Removing hair pixels at test time only closes 44–47% of the ERM clean–reversed gap: most of what a frozen encoder exploits in a real hair trap is not the hair pixels but what accompanies hair (hair-bearing images differ in sex, age and body site). This bounds every pixel-level method, including oracle inpainting and U-MtE, and explains why image-level-label methods, which remove everything correlated with the label, recover more.

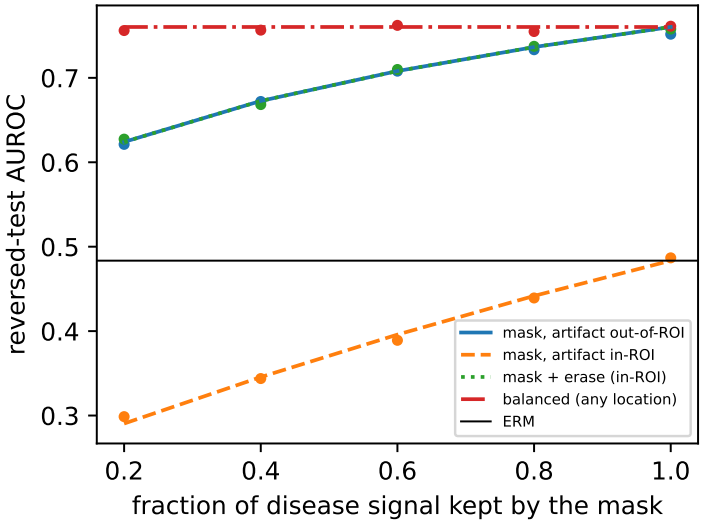


Figure 3: Reversed-test AUROC predicted by the linear-Gaussian model (lines) and simulated with logistic heads (points) as masking removes a growing share of disease signal ($S = 1$, $A = 2$, 90/10 training). Generated by `scripts/analysis/theory_sim.py`.

The linear-Gaussian account: simulation check.

Does the account match real encoders?. For any test composition $q_y = P(A{=}1 \mid Y{=}y)$ the same head gives $\mathrm{AUROC} = \Phi\big(\Delta/\sqrt{\sigma_1^2 + \sigma_0^2}\big)$ with $\Delta = S + (p_1 - p_0)(q_1 - q_0)A/(1 + vA)$, $\sigma_y^2 = S + w_a^2\,(1 + Aq_y(1 - q_y))$, $w_a = \sqrt{A}(p_1 - p_0)/(1 + vA)$ and $v = [p_1(1 - p_1) + p_0(1 - p_0)]/2$. In a pre-registered test we fitted (S, A) for every trained model to its clean and correlated AUROC *only* and compared the implied reversed-test

AUROC with the held-out observation; no parameter is shared across models or tuned (main-text figure of the theory test). Two reference predictors use the same two inputs: "reversed = clean" (no shortcut harm) and a linear-symmetry heuristic, reversed = 2·clean − correlated. The endpoint on which the account is most informative is the held-out reversed-test AUROC itself: over 136 ERM and masking models (real traps and controlled sweeps) its mean absolute error is 0.052 AUROC ($r = 0.91$), against 0.129 for the symmetry heuristic and 0.237 for "reversed = clean". Along the controlled overlap sweeps it recovers the sign of the masking gain at 44 of 50 overlap points (symmetry heuristic: 38/50). For the location crossover the gain over the heuristic is small: both follow the observed crossovers closely across nine cohort–backbone pairs (account: mean absolute error 0.015, 9/9 signs; heuristic: error 0.044, 9/9), because a crossover is a difference of masking gains in which most of the symmetry already cancels. We therefore do not use the crossover correlation as evidence for the model. The account also matches end-to-end fine-tuned networks (3/3 crossover signs), which violate its assumptions, and, fitted to DermLIP's clean and correlated AUROC, it anticipates that encoder's absence of a hair location gap (+0.013 implied, +0.014 observed; S18). Adding the chest-radiograph cells (device traps and drains, RAD-DINO; 154 models) leaves the error almost unchanged (0.053, $r = 0.88$), but one near-zero device crossover disagrees in sign (consolidation: observed −0.004, implied +0.008; 12 of 13 signs), which fails our pre-registered all-signs criterion, so we call the model a qualitative account. Its one clear failure is the drain cohort (implied reversed AUROC 0.72–0.78, observed 0.19–0.23): the clean and correlated environments do not expose a shortcut carried by correlates of the drain, regime (v).

S15. Per-seed versus crossed bootstrap

The original estimator resamples training-seed clusters and then test images independently within each sampled cluster. When the clusters share test images—every seed of an analysis scores the same official thyroid test split, the same transplant recipients, and largely the same trap pool—this treats one test set as several and understates the test-set sampling variance. The crossed estimator resamples clusters and draws one Poisson weight per test image, shared by every cluster (`wtss.stats.crossed_auc_bootstrap`). The table gives the crossed CI of each headline claim, its width relative to the per-seed CI of the same claim and whether the "excludes zero" verdict changed; the per-seed CIs themselves, and the same comparison for all 2,466 intervals of the regenerated runs (median width ratio 1.56, IQR 1.34–1.74; 179 verdicts changed) and for the 263 intervals of the four analyses regenerated later (LaMa, template erasure, chest drains, chest devices; 32 verdicts changed), are listed in `results/bootstrap_correction/`. None of the four changed verdicts below carries a claim of the paper. Every interval in the paper and this supplement is a crossed one; the exceptions (image bootstrap for label agreement, point estimates for runs without per-image predictions) are listed in the protocol section of the main text. Generated by `scripts/analysis/crossed_ci.py`.

Claim	Cohort / test set	Crossed seed×image bootstrap	Width / per-seed width	Verdict changed
real-trap crossover (repro)	ISIC hair	$+0.147$ $[+0.109, +0.185]$	1.31	no
real-trap crossover (matched)	ISIC hair	$+0.154$ $[+0.122, +0.187]$	1.12	no
real-trap crossover (matched05)	ISIC hair	$+0.124$ $[+0.086, +0.162]$	1.15	no
real-trap crossover (repro)	Thyroid	$+0.239$ $[+0.196, +0.280]$	1.61	no
real-trap crossover (matched)	Thyroid	$+0.238$ $[+0.186, +0.291]$	1.70	no
real-trap crossover (matched05)	Thyroid	$+0.218$ $[+0.165, +0.270]$	1.65	no
real-trap crossover (repro)	Ovary	$+0.170$ $[+0.087, +0.253]$	1.55	no
real-trap crossover (matched)	Ovary	$+0.163$ $[+0.049, +0.274]$	1.50	no
real-trap crossover (matched05)	Ovary	$+0.152$ $[+0.041, +0.265]$	1.41	no
real-trap crossover (repro)	Capsule	$+0.377$ $[+0.326, +0.425]$	1.36	no
real-trap crossover (matched)	Capsule	$+0.230$ $[+0.029, +0.422]$	1.86	no
real-trap crossover (matched05)	Capsule	$+0.328$ $[+0.087, +0.573]$	1.74	no
transplant interaction (artifact)	ISIC hair	$+0.253$ $[+0.214, +0.293]$	1.76	no
transplant interaction (balanced)	ISIC hair	-0.052 $[-0.071, -0.033]$	1.43	no
transplant interaction (neutral paste)	ISIC hair	$+0.204$ $[+0.164, +0.244]$	1.71	no
transplant interaction (artifact)	Thyroid	$+0.277$ $[+0.256, +0.298]$	1.63	no
transplant interaction (balanced)	Thyroid	-0.016 $[-0.025, -0.007]$	1.57	no
transplant interaction (neutral paste)	Thyroid	$+0.145$ $[+0.130, +0.161]$	1.99	no
transplant interaction (artifact)	Ovary	$+0.074$ $[+0.044, +0.104]$	1.61	no
transplant interaction (balanced)	Ovary	-0.011 $[-0.025, +0.002]^*$	1.52	yes
transplant interaction (neutral paste)	Ovary	$+0.052$ $[+0.021, +0.084]$	1.95	no
transplant interaction (artifact)	Capsule	$+0.497$ $[+0.427, +0.567]$	1.68	no
transplant interaction (balanced)	Capsule	-0.032 $[-0.068, +0.004]^*$	1.60	yes
transplant interaction (neutral paste)	Capsule	$+0.459$ $[+0.383, +0.535]$	1.57	no
natural mte - mask (all pairs)	Thyroid (official split)	$+0.004$ $[-0.012, +0.020]^*$	1.77	no
natural mte_protect - mask (all pairs)	Thyroid (official split)	-0.021 $[-0.034, -0.008]$	1.90	no
natural balanced - mask (all pairs)	Thyroid (official split)	-0.017 $[-0.064, +0.030]^*$	2.03	no
natural mte_balanced - mask (all pairs)	Thyroid (official split)	-0.005 $[-0.023, +0.013]^*$	1.86	no
natural mask - ERM (hard pairs)	Thyroid (official split)	-0.095 $[-0.167, -0.023]$	2.14	no
natural mask - ERM (all pairs)	Thyroid (official split)	-0.004 $[-0.048, +0.041]^*$	2.13	no
natural mte - mask (all pairs)	ISIC, BCN held out	$+0.007$ $[+0.004, +0.010]$	1.58	no
natural mte_protect - mask (all pairs)	ISIC, BCN held out	-0.019 $[-0.024, -0.014]$	1.57	no
natural balanced - mask (all pairs)	ISIC, BCN held out	$+0.022$ $[+0.012, +0.031]$	2.06	no
natural mte_balanced - mask (all pairs)	ISIC, BCN held out	$+0.010$ $[+0.006, +0.015]$	1.67	no
natural mask - ERM (hard pairs)	ISIC, BCN held out	-0.025 $[-0.042, -0.008]$	1.51	no
natural mask - ERM (all pairs)	ISIC, BCN held out	-0.025 $[-0.035, -0.016]$	2.05	no
natural mte - mask (all pairs)	ISIC, HAM held out	$+0.013$ $[+0.009, +0.017]$	1.67	no
natural mte_protect - mask (all pairs)	ISIC, HAM held out	-0.006 $[-0.016, +0.004]^*$	1.18	no
natural balanced - mask (all pairs)	ISIC, HAM held out	-0.032 $[-0.049, -0.013]$	1.38	no
natural mte_balanced - mask (all pairs)	ISIC, HAM held out	$+0.018$ $[+0.009, +0.028]$	1.15	no
natural mask - ERM (hard pairs)	ISIC, HAM held out	$+0.050$ $[+0.030, +0.070]$	2.21	no
natural mask - ERM (all pairs)	ISIC, HAM held out	$+0.031$ $[+0.017, +0.045]$	2.13	no
natural mte - mask (all pairs)	ISIC, MSK held out	-0.008 $[-0.014, -0.002]$	2.04	no
natural mte_protect - mask (all pairs)	ISIC, MSK held out	-0.003 $[-0.012, +0.006]^*$	1.85	no
natural balanced - mask (all pairs)	ISIC, MSK held out	-0.018 $[-0.041, +0.004]^*$	2.04	yes
natural mte_balanced - mask (all pairs)	ISIC, MSK held out	-0.016 $[-0.024, -0.008]$	1.64	no
natural mask - ERM (hard pairs)	ISIC, MSK held out	-0.065 $[-0.111, -0.020]$	1.72	no
natural mask - ERM (all pairs)	ISIC, MSK held out	$+0.016$ $[-0.006, +0.038]^*$	1.91	yes
natural mte - mask (all pairs)	Capsule (held-out frames)	-0.005 $[-0.011, +0.001]^*$	1.18	no
natural mte_protect - mask (all pairs)	Capsule (held-out frames)	-0.023 $[-0.029, -0.017]$	1.41	no
natural balanced - mask (all pairs)	Capsule (held-out frames)	-0.086 $[-0.109, -0.065]$	1.13	no
natural mte_balanced - mask (all pairs)	Capsule (held-out frames)	-0.008 $[-0.013, -0.001]$	1.19	no
natural mask - ERM (hard pairs)	Capsule (held-out frames)	$+0.021$ $[-0.003, +0.048]^*$	1.16	no
natural mask - ERM (all pairs)	Capsule (held-out frames)	$+0.077$ $[+0.059, +0.097]$	1.21	no
fine-tuned crossover (ResNet-50)	ISIC hair	$+0.143$ $[+0.056, +0.231]$	1.11	no

*Crossed CI includes zero. The original intervals are listed in `results/bootstrap_correction/sweeps_umte_old_vs_new.md`.

S16. Later registered analyses: implementations, overlap, external natural test, labels, fine-tuning, thresholds

Four registrations added after the main analyses (`PREREGISTRATION_ROUND4` to `ROUND7`; registry in S1), each committed and pushed before its analysis; the hypothesis identifiers below (FT0–FT4, TM1–TM2) are those of the registrations. Full reports: `stages/stage7_robustness_external` and `stages/stage8_labels_strong_model`.

Masking implementations and the real overlap. Five views per image from the cached 518-px image and its ROI: mean-colour fill (the main analysis), black fill, a 16-px Gaussian blur of everything outside the ROI, a crop to the ROI's bounding box, and the same crop of the masked image. For the dose-response, all artifact-bearing images of a cohort are binned by r (registered bins, merged by a registered rule when a bin is too small); the slope of the mask gain on the mean r of each bin is estimated with the crossed bootstrap.

Table 8: Location crossover C_v for every masking implementation (DINOv2 ViT-B/14, crossed 95% CIs). All 16 non-reference crossovers are positive with Holm-adjusted $p \leq 0.010$ (family of 16).

Cohort	mean-colour fill (reference)	black fill	blurred background	box crop	box crop of the masked image
Dermoscopy hair (ISIC 2019)	+0.147 [+0.111, +0.185]	+0.170 [+0.133, +0.207]	+0.154 [+0.117, +0.191]	+0.091 [+0.064, +0.117]	+0.154 [+0.121, +0.187]
Thyroid calipers	+0.239 [+0.196, +0.280]	+0.223 [+0.181, +0.265]	+0.218 [+0.177, +0.258]	+0.246 [+0.206, +0.285]	+0.250 [+0.210, +0.290]
Capsule debris	+0.377 [+0.328, +0.426]	+0.367 [+0.318, +0.416]	+0.278 [+0.221, +0.330]	+0.317 [+0.267, +0.367]	+0.358 [+0.305, +0.409]
Ovary calipers	+0.170 [+0.088, +0.250]	+0.153 [+0.063, +0.242]	+0.112 [+0.027, +0.195]	+0.145 [+0.083, +0.207]	+0.160 [+0.080, +0.240]

Table 9: Secondary contrast $C_{\text{mask}} - C_v$: how much smaller the crossover of an implementation is than that of mean-colour fill (positive = weaker location effect than the reference).

Cohort	black fill	blurred background	box crop	box crop of the masked image
Dermoscopy hair (ISIC 2019)	−0.023 [−0.040, −0.007]	−0.006 [−0.042, +0.024]	+0.056 [+0.012, +0.099]	−0.007 [−0.033, +0.019]
Thyroid calipers	+0.016 [−0.004, +0.036]	+0.020 [−0.008, +0.049]	−0.007 [−0.049, +0.032]	−0.011 [−0.037, +0.014]
Capsule debris	+0.010 [−0.001, +0.021]	+0.099 [+0.067, +0.132]	+0.059 [+0.032, +0.087]	+0.019 [+0.001, +0.037]
Ovary calipers	+0.017 [−0.034, +0.079]	+0.058 [+0.004, +0.113]	+0.024 [−0.045, +0.096]	+0.010 [−0.046, +0.068]

Table 10: Dose-response: least-squares slope of $g_b = \text{rev(mask)} - \text{rev(ERM)}$ on the bin mean overlap r (crossed 95% CIs; the artifact-free images shared by all bins carry one bootstrap weight).

Cohort	Bins	Slope of the mask gain on r	Holm p	Spearman	Zero crossing r	H9
Dermoscopy hair (ISIC 2019)	5	−0.235 [−0.276, −0.194]	< 0.001	-1.00	0.401	supported
Thyroid calipers	5	−0.250 [−0.303, −0.197]	< 0.001	-1.00	none	supported
Capsule debris	5	−0.489 [−0.539, −0.438]	< 0.001	-1.00	none	supported
Ovary calipers	3	−0.205 [−0.300, −0.109]	< 0.001	-1.00	none	supported

Prospective test of the linear-Gaussian account. The procedure of the main text (linear-Gaussian account), unchanged, applied to cells that did not exist when it was registered.

Table 11: Prospective theory test: fitted to each cell's clean and correlated AUROC only; the reversed AUROC and every quantity derived from it are held out. None of these cells existed when the rule was registered.

Endpoint	Cells	Theory	"reversed = clean"	Symmetric reversal
Reversed AUROC, all new cells	84	MAE 0.011, $r = 0.996$	MAE 0.177	MAE 0.036
Masking-implementation crossovers	16	signs 16/16, $r = 0.992$, MAE 0.011		
Overlap dose-response slopes	4	signs 4/4, MAE 0.018		
Overlap-bin gains	18	signs 18/18, MAE 0.014		

ISIC 2019 → ISIC 2020 at natural prevalence. All ISIC 2019 images for training and validation; every ISIC 2020 image with a lesion mask for testing, with near-duplicates of ISIC 2019 images removed. The registered rule (perceptual-hash distance ≤ 8) removed 23% of ISIC 2020, mostly look-alike images of other patients; a label-free calibrated rule (hash and embedding thresholds set from the false-match rate between different patients) was registered as a sensitivity analysis before any model of it was fitted.

Table 12: ISIC 2020 under four duplicate rules, from one set of predictions for all 32,997 images (crossed 95% CIs). The calibrated rule is label-free and was committed before any model of this analysis was fitted.

Duplicate rule	Removed	Test images	Melanomas	mask − ERM, all pairs	hard pairs	easy pairs	balanced − mask, hard pairs
exact copies only	26	32,971	578	$+0.017$ $[-0.000, +0.033]$	-0.037 $[-0.061, -0.012]$	$+0.048$ $[+0.009, +0.090]$	$+0.072$ $[+0.048, +0.096]$
hash distance ≤ 2	138	32,859	575	$+0.016$ $[-0.000, +0.032]$	-0.037 $[-0.062, -0.013]$	$+0.048$ $[+0.008, +0.090]$	$+0.072$ $[+0.048, +0.097]$
calibrated (primary)	45	32,952	575	$+0.016$ $[-0.000, +0.033]$	-0.037 $[-0.061, -0.012]$	$+0.048$ $[+0.009, +0.090]$	$+0.072$ $[+0.048, +0.097]$
hash distance ≤ 8 (registered)	7,657	25,340	483	$+0.020$ $[+0.003, +0.038]$	-0.026 $[-0.052, -0.000]$	$+0.045$ $[+0.005, +0.087]$	$+0.059$ $[+0.035, +0.084]$

Table 13: Cleaner hair groups on the calibrated ISIC 2020 test set: hair-free ($\leq \tau$ predicted hair pixels), clear in-lesion hair ($\geq 3\tau$, $r \geq 0.5$) and clear hair beside the lesion only ($\geq 3\tau$, $r < 0.1$); ambiguous images excluded from these pairs.

Positives	Negatives	Contrast	AUROC difference [95% CI]	Status
hair-free melanomas	benign, clear in-lesion hair	mask − ERM	-0.005 $[-0.057, +0.048]$	H13
melanomas, clear in-lesion hair	hair-free benign	mask − ERM	$+0.022$ $[-0.019, +0.064]$	descriptive
melanomas, hair beside the lesion only	hair-free benign	mask − ERM	-0.028 $[-0.060, +0.004]$	descriptive
hair-free melanomas	benign, hair beside the lesion only	mask − ERM	$+0.059$ $[+0.010, +0.112]$	descriptive
hair-free melanomas	benign, clear in-lesion hair	balanced − mask	$+0.016$ $[-0.039, +0.075]$	descriptive

Artifact labels checked without a clinician. Sources, licences and coverage were recorded before any agreement was computed; images on which our own labellers were trained are excluded. A model labeller counts only if its κ is ≥ 0.40 on the images on which the two other caliper labels agree; the annotation detector is declared invalid if it fires on more than half of our caliper-free images. An image is removed if an informative source places it in another cell (artifact present vs absent, or the other side of the boundary); images without a source are kept.

Table 14: Independent label sources and their coverage of our cohorts after the registered exclusions (training images of our own labellers). No image of the thyroid or ovary cohorts left the GPU machine: BUSClean and MedGemma ran locally.

Cohort	Source	Used for	Licence	Matched images	Gate
Dermoscopy hair	DermArtifactDB	hair present / absent	CC BY 4.0	20,519	analysed
Dermoscopy hair	IMA++	lesion masks, 1–5 annotators	cc-by-nc-nd-4.0	241	analysed
Dermoscopy hair	Kabir hair masks	hair masks	CC BY 4.0	485	analysed
Capsule debris	Multicentre clear/contaminated capsule masks	contamination masks	CC BY 4.0	29	not feasible
Thyroid calipers	BUSClean	annotation detector (`detect_anno`)	MIT	3,493	analysed
Ovary calipers	BUSClean	annotation detector (`detect_anno`)	MIT	1,202	analysed
Thyroid calipers	MedGemma 1.5 4B-it	vision–language answers (presence, location)	model terms of use	3,492	analysed
Ovary calipers	MedGemma 1.5 4B-it	vision–language answers (presence, location)	model terms of use	1,198	analysed

Table 15: Dermoscopy: agreement of our labels with three independent sources (95% bootstrap CIs over images, 2,000 replicates). The Kabir images are all hair images and our masks mark hair on every one of them, so κ is not informative there; the raw agreement is given instead.

Source	Quantity	Images	All	$Y = 0$ (benign)	$Y = 1$ (melanoma)
DermArtifactDB	hair present (ours: >30 hair pixels)	20,519	κ 0.076 [0.070, 0.082]	κ 0.083 [0.076, 0.090]	κ 0.051 [0.041, 0.060]
IMA++ (majority mask)	lesion mask (Dice)	241	0.851 [0.830, 0.870]	0.851 [0.825, 0.872]	0.853 [0.819, 0.888]
IMA++	trap class (r recomputed)	239	0.841 [0.791, 0.887]	0.855 [0.807, 0.899]	0.750 [0.594, 0.875]
Kabir et al.	hair mask (Dice)	485	0.618 [0.600, 0.637]	0.617 [0.595, 0.638]	0.629 [0.592, 0.665]
Kabir et al.	hair present	485	agreement 97.3%		

Table 16: Thyroid: how often each independent labeller reports a caliper in each of our cells, and the share of images the registered rule removes (an image is removed if *any* informative labeller places it in another cell).

Our cell	Images	BUSClean: annotation found	MedGemma: caliper "yes"	Contradicted (registered rule)
caliper-free	1,801	5%	29%	30%
Trap A (in the nodule)	1,150	56%	99%	61%
Trap B (outside)	326	37%	87%	87%

Table 17: Thyroid, **exploratory**: the images on which BUSClean and MedGemma give the same cell. An image counts as contradicted only if both place it in the same cell and that cell differs from ours.

Our cell	Both labellers agree with each other	... and with our cell	Contradicted by both	benign	malignant
caliper-free	1,301	97%	2%	2%	2%
Trap A (in the nodule)	475	94%	2%	3%	1%
Trap B (outside)	89	48%	14%	11%	21%

Table 18: Traps re-run on the images no informative source contradicts (DINOv2 ViT-B/14, protocol of the real traps, crossed 95% CIs, Holm over the cohorts that ran). For capsule and ovary no image was removed, so the re-run reproduces the original analysis and adds no evidence about the labels.

Cohort	Original crossover	Uncontradicted labels	Removed (Trap A / Trap B)	Registered test
Dermoscopy hair (ISIC 2019)	+0.147 [+0.111, +0.185]	+0.199 [+0.164, +0.233]	42% / 43%	supported
Thyroid calipers	+0.239 [+0.196, +0.280]	—	61% / 87%	count gate not met (Trap B)
Capsule debris	+0.377 [+0.327, +0.426]	+0.377 [+0.327, +0.425]	0% / 0%	supported; nothing removed (no source analysable)
Ovary calipers	+0.170 [+0.087, +0.252]	+0.170 [+0.089, +0.255]	0% / 0%	supported; nothing removed (no informative source yet)

Fine-tuned thyroid model. Twelve recipes compared on validation images only; the chosen recipe was committed before the final models were trained.

Table 19: Recipe selection on validation images only; no test image was scored. The chosen recipe (bold) was committed before any final model was trained.

Architecture	Learning rate	Epochs	Validation AUROC (ERM, seed 42)
ConvNeXt-T (IN-22k)	3e-5	8	0.849
ConvNeXt-T (IN-22k)	3e-5	20	0.842
ConvNeXt-T (IN-22k)	1e-4	8	**0.865**
ConvNeXt-T (IN-22k)	1e-4	20	0.865
ResNet-50	3e-5	8	0.658
ResNet-50	3e-5	20	0.723
ResNet-50	1e-4	8	0.723
ResNet-50	1e-4	20	0.781
DINOv2 ViT-B/14	3e-5	8	0.838
DINOv2 ViT-B/14	3e-5	20	0.850
DINOv2 ViT-B/14	1e-4	8	0.715
DINOv2 ViT-B/14	1e-4	20	0.804

Table 20: Part B: the chosen fine-tuned model on the official patient-disjoint thyroid split (5 seeds × 4 arms) and in the thyroid traps (10 bootstrap clusters). Crossed bootstrap; Holm over FT1–FT4.

	Hypothesis	Estimate [95% CI]	p (one-sided)	Holm p	Verdict
FT0	ERM test AUROC ≥ 0.773 (descriptive)	0.769	—	—	below benchmark
FT1	hard-pair AUROC, mask − ERM < 0	-0.102 $[-0.178, -0.026]$	0.0040	0.0040	supported
FT2	sensitivity loss (CI < 0) at ≥ 3 of OP2–OP5	4 of 4	0.0005	0.0015	supported
FT3	OP2 sensitivity, malignant with in-ROI caliper, mask − ERM < 0	-0.521 $[-0.642, -0.382]$	0.0005	0.0015	supported
FT4	thyroid trap crossover > 0	$+0.187$ $[+0.111, +0.268]$	0.0001	0.0004	supported

Table 21: Fine-tuned model, official patient-disjoint thyroid test split: masking at five validation-fixed operating points (crossed bootstrap, thresholds re-estimated in every replicate).

Operating point (fixed on validation)	Sensitivity ERM → mask	Δ sensitivity	Δ specificity	Sensitivity, malignant with in-ROI caliper (ERM → mask)	Δ in that subgroup
OP1 max. balanced accuracy	0.781 → 0.408	-0.373 $[-0.508, -0.243]$	$+0.239$ $[+0.155, +0.345]$	0.779 → 0.269	-0.510 $[-0.660, -0.346]$
OP2 specificity ≥ 0.80	0.747 → 0.371	-0.376 $[-0.462, -0.297]$	$+0.225$ $[+0.163, +0.291]$	0.744 → 0.223	-0.521 $[-0.642, -0.382]$
OP3 specificity ≥ 0.90	0.542 → 0.218	-0.324 $[-0.419, -0.228]$	$+0.143$ $[+0.098, +0.198]$	0.526 → 0.108	-0.418 $[-0.536, -0.276]$
OP4 sensitivity ≥ 0.80	0.783 → 0.423	-0.360 $[-0.439, -0.257]$	$+0.232$ $[+0.159, +0.302]$	0.779 → 0.267	-0.513 $[-0.627, -0.371]$
OP5 sensitivity ≥ 0.90	0.888 → 0.646	-0.242 $[-0.345, -0.162]$	$+0.240$ $[+0.159, +0.345]$	0.867 → 0.510	-0.356 $[-0.497, -0.225]$

Thresholds matched on the test set. Each arm's threshold is chosen on the test images themselves, the best case for any re-tuning at a deployment site; the crossed bootstrap re-estimates the thresholds in every replicate.

Table 22: Sensitivity when each arm's threshold is chosen on the test set itself, so that both arms reach the same test specificity (or sensitivity); ERM → mask, and mask − ERM with crossed 95% CIs. The first row of each model is the registered test (TM1, TM2).

Model	Both arms matched at	Malignant, in-ROI caliper (78)	Malignant, no in-ROI caliper	All malignant	Specificity ERM / mask
fine-tuned	test specificity 0.80	0.544 → 0.390, -0.154 $[-0.302, -0.023]$	0.563 → 0.614, $+0.051$ $[-0.070, +0.150]$	0.557 → 0.540, -0.017 $[-0.124, +0.070]$	0.806 / 0.804
	test specificity 0.90	0.323 → 0.218, -0.105 $[-0.221, +0.019]$	0.343 → 0.437, $+0.094$ $[-0.028, +0.196]$	0.336 → 0.364, $+0.028$ $[-0.070, +0.119]$	0.903 / 0.904
	test sensitivity 0.80	0.803 → 0.715, -0.087 $[-0.172, +0.018]$	0.800 → 0.843, $+0.043$ $[-0.009, +0.083]$	0.801 → 0.801, $+0.000$ $[-0.005, +0.006]$	0.611 / 0.542
frozen probe	test specificity 0.80	0.505 → 0.331, -0.174 $[-0.313, -0.012]$	0.514 → 0.638, $+0.124$ $[+0.001, +0.231]$	0.511 → 0.536, $+0.025$ $[-0.083, +0.132]$	0.803 / 0.803
	test specificity 0.90	0.231 → 0.151, -0.079 $[-0.244, +0.086]$	0.313 → 0.390, $+0.077$ $[-0.040, +0.200]$	0.286 → 0.311, $+0.025$ $[-0.085, +0.141]$	0.903 / 0.904
	test sensitivity 0.80	0.779 → 0.713, -0.067 $[-0.155, +0.026]$	0.811 → 0.844, $+0.033$ $[-0.013, +0.075]$	0.801 → 0.801, $+0.000$ $[-0.006, +0.006]$	0.538 / 0.515

Table 23: Descriptive: the other unaltered test sets at matched test specificity 0.80, mask − ERM (crossed 95% CIs); conflicting = positives whose artifact status conflicts with the test set's own association.

Test set (frozen probe)	Conflicting positives	Aligned positives	All positives
Dermoscopy, BCN held out	-0.040 $[-0.065, -0.015]$	-0.034 $[-0.062, -0.003]$	-0.038 $[-0.057, -0.017]$
Dermoscopy, HAM held out	$+0.067$ $[+0.032, +0.106]$	$+0.039$ $[-0.019, +0.095]$	$+0.059$ $[+0.027, +0.093]$
Dermoscopy, MSK held out	-0.001 $[-0.054, +0.053]$	$+0.193$ $[+0.067, +0.311]$	$+0.020$ $[-0.030, +0.070]$
Capsule (held-out frames)	$+0.037$ $[-0.013, +0.110]$	$+0.144$ $[+0.107, +0.196]$	$+0.131$ $[+0.098, +0.179]$

Table 24: Every remedy against masking on the confirmation data of the first registered round: arm − masking with crossed 95% CIs (the main-text remedies table with its intervals).

Test	ERM (no mask)	Group balancing	U-MtE	U-MtE + balancing	Class-conditional constraint
Needs artifact labels	no	yes	no	yes	yes
Trap A: artifact inside the ROI					
Thyroid	−0.111 [−0.134, −0.087]	+0.270 [+0.235, +0.306]	+0.224 [+0.187, +0.259]	+0.393 [+0.369, +0.414]	+0.330 [+0.311, +0.351]
Capsule	−0.081 [−0.111, −0.051]	+0.099 [+0.071, +0.128]	−0.119 [−0.141, −0.098]	+0.187 [+0.159, +0.216]	+0.153 [+0.131, +0.177]
Dermoscopy hair	+0.054 [+0.023, +0.087]	+0.286 [+0.246, +0.326]	+0.041 [+0.028, +0.054]	+0.228 [+0.200, +0.256]	+0.193 [+0.166, +0.221]
Ovary*	−0.089 [−0.139, −0.037]	+0.178 [+0.124, +0.228]	+0.068 [+0.042, +0.093]	+0.213 [+0.178, +0.247]	+0.118 [+0.089, +0.147]
Trap B: artifact outside the ROI					
Thyroid	−0.335 [−0.377, −0.292]	−0.109 [−0.166, −0.071]	−0.000 [−0.017, +0.017]	+0.026 [−0.021, +0.047]	+0.015 [−0.000, +0.026]
Capsule	−0.468 [−0.509, −0.428]	−0.224 [−0.264, −0.185]	−0.020 [−0.034, −0.006]	+0.004 [−0.019, +0.024]	+0.022 [+0.014, +0.030]
Dermoscopy hair	−0.115 [−0.156, −0.074]	+0.097 [+0.061, +0.133]	+0.010 [−0.012, +0.031]	+0.120 [+0.077, +0.141]	+0.045 [+0.033, +0.058]
Ovary*	−0.274 [−0.348, −0.196]	−0.037 [−0.103, +0.028]	+0.021 [−0.011, +0.055]	+0.037 [−0.024, +0.073]	−0.002 [−0.017, +0.012]
Unaltered test sets, all pairs					
Thyroid (official split)	+0.009 [−0.035, +0.053]	−0.009 [−0.055, +0.039]	+0.009 [−0.006, +0.025]	+0.001 [−0.015, +0.018]	−0.002 [−0.011, +0.007]
Capsule	−0.065 [−0.079, −0.050]	−0.068 [−0.087, −0.052]	−0.007 [−0.014, −0.002]	−0.008 [−0.013, −0.004]	−0.000 [−0.001, +0.001]
ISIC, BCN held out	+0.027 [+0.017, +0.037]	+0.023 [+0.013, +0.033]	+0.005 [+0.001, +0.009]	+0.009 [+0.004, +0.015]	−0.005 [−0.006, −0.003]
ISIC, HAM held out	−0.037 [−0.052, −0.022]	−0.041 [−0.057, −0.026]	+0.010 [+0.004, +0.017]	+0.016 [+0.006, +0.025]	−0.003 [−0.008, +0.002]
ISIC, MSK held out	−0.015 [−0.037, +0.008]	−0.017 [−0.040, +0.005]	−0.006 [−0.012, +0.000]	−0.013 [−0.022, −0.005]	−0.002 [−0.006, +0.002]
ISIC 2019 → 2020	−0.012 [−0.031, +0.007]	−0.025 [−0.046, −0.006]	−0.016 [−0.021, −0.010]	−0.035 [−0.043, −0.027]	−0.021 [−0.027, −0.016]
Unaltered test sets, conflicting pairs					
Thyroid (official split)	+0.092 [+0.017, +0.168]	+0.140 [+0.066, +0.214]	+0.035 [+0.004, +0.067]	+0.075 [+0.043, +0.110]	+0.084 [+0.069, +0.101]
ISIC, BCN held out	+0.035 [+0.017, +0.051]	+0.054 [+0.039, +0.070]	+0.003 [−0.004, +0.011]	+0.038 [+0.029, +0.046]	+0.036 [+0.033, +0.040]
ISIC, MSK held out	+0.065 [+0.023, +0.108]	+0.102 [+0.062, +0.142]	−0.010 [−0.022, +0.003]	+0.029 [+0.007, +0.049]	+0.070 [+0.059, +0.082]
ISIC 2019 → 2020	+0.055 [+0.033, +0.078]	+0.087 [+0.064, +0.109]	+0.003 [−0.005, +0.011]	+0.043 [+0.030, +0.054]	+0.078 [+0.069, +0.088]

S17. Is there a remedy that dominates masking?

Two registrations (`PREREGISTRATION_ROUND8` and `ROUND9`; registry in S1) and a documented search between them (`docs/ROUND9_SEARCH_LEDGER.md`). Candidates are heads on frozen DINOv2-B/14 features of the masked image (or the full image); every setting is chosen per training set on validation data by the worst-group validation AUROC, subject to the same-artifact validation AUROC staying within 0.005 of masking's, with masking as fallback. A candidate dominates if every component of the criterion holds (intersection–union test, one-sided 0.025 each); across candidates a fixed sequence registered in advance. Confirmation: trap resamples generated only after each registration (seeds 8101–8505 and 9101–9505), ovary held out from every choice; the unaltered test sets and ISIC 2020 had been seen before and are semi-confirmatory. Full reports: `stages/stage9_remedies_dominance`.

Table 25: The registered criterion for "dominates masking". A candidate dominates only if every component is met; easy pairs, where any method that stops using the shortcut must lose, are not in it.

	Cells	Test (candidate − masking, one-sided 0.025)
D1	all pairs of the six unaltered test sets (thyroid split, capsule, BCN, HAM, MSK, ISIC 2019 → 2020)	not worse: lower bound > -0.01
D2	shortcut-conflicting pairs (thyroid, BCN, MSK, ISIC 2020) and Trap A min(reversed, correlated) in four cohorts	better: lower bound > 0
D3	Trap B and clean tests, per cell (16) and pooled over cohorts (4)	not worse: -0.03 per cell, -0.01 pooled
D4	every trap	no shortcut flipping: correlated $\geq$ reversed -0.02

Table 26: First registered round, on its confirmation data: no candidate dominates masking. The sequence stopped at its first candidate; the others are reported unadjusted.

Candidate	Components met	All of D2 met	Losses	Fixed sequence
class-conditional mean constraint (`mask_cmc`)	40 of 42	yes	ISIC 2019 → 2020 (D1)	tested, not rejected
mean constraint on the full image (`full_cmc`)	36 of 42	no	dermoscopy, HAM held out (D1), dermoscopy, MSK held out (D1), ISIC 2019 → 2020 (D1)	not tested in the sequence
partial reweighting (`mask_bal`)	39 of 42	yes	thyroid split (D1), dermoscopy, MSK held out (D1), ISIC 2019 → 2020 (D1)	not tested in the sequence
location-matched pasting (`locrand_loc`)	39 of 42	no	dermoscopy, BCN held out (D2)	not tested in the sequence

Table 27: First registered round: the two candidates that won everywhere masking fails: the D1 and D2 components and the pooled D3 components (AUROC, crossed 95% CIs); every per-cell D3 component and D4 were met by both.

Component	Cell	`mask_cmc` − masking		`mask_bal` − masking	
D1 all pairs	thyroid split	−0.002	inconclusive	−0.008	**loss**
		[−0.011, +0.007]		[−0.017, +0.001]	
D1 all pairs	capsule	−0.000	met	−0.001	met
		[−0.001, +0.001]		[−0.004, +0.000]	
D1 all pairs	dermoscopy, BCN held out	−0.005	met	+0.003	met
		[−0.006, −0.003]		[+0.000, +0.007]	
D1 all pairs	dermoscopy, HAM held out	−0.003	met	+0.003	met
		[−0.008, +0.002]		[−0.001, +0.008]	
D1 all pairs	dermoscopy, MSK held out	−0.002	met	−0.006	**loss**
		[−0.006, +0.002]		[−0.011, −0.000]	
D1 all pairs	ISIC 2019 → 2020	−0.021	**loss**	−0.011	**loss**
		[−0.027, −0.016]		[−0.016, −0.005]	
D2 conflicting pairs	thyroid split	+0.084	met	+0.048	met
		[+0.069, +0.101]		[+0.032, +0.066]	
D2 conflicting pairs	dermoscopy, BCN held out	+0.036	met	+0.033	met
		[+0.033, +0.040]		[+0.026, +0.040]	
D2 conflicting pairs	dermoscopy, MSK held out	+0.070	met	+0.044	met
		[+0.059, +0.082]		[+0.030, +0.058]	
D2 conflicting pairs	ISIC 2019 → 2020	+0.078	met	+0.035	met
		[+0.069, +0.088]		[+0.024, +0.046]	
D2 Trap A	thyroid	+0.330	met	+0.380	met
		[+0.311, +0.351]		[+0.355, +0.404]	
D2 Trap A	capsule	+0.153	met	+0.166	met
		[+0.131, +0.177]		[+0.148, +0.185]	
D2 Trap A	ovary (held out)	+0.118	met	+0.161	met
		[+0.089, +0.147]		[+0.135, +0.187]	
D2 Trap A	dermoscopy hair	+0.193	met	+0.213	met
		[+0.166, +0.221]		[+0.187, +0.239]	
D3 pooled	Trap B reversed	+0.020	met	+0.045	met
		[+0.014, +0.026]		[+0.036, +0.054]	
D3 pooled	clean (Trap A models)	+0.057	met	+0.063	met
		[+0.048, +0.065]		[+0.054, +0.072]	
D3 pooled	clean (Trap B models)	+0.007	met	+0.010	met
		[+0.002, +0.012]		[+0.002, +0.017]	

Table 28: The exploratory search: twelve candidates on development data only (development seeds, validation folds of the unaltered training sets; never a test set used for confirmation, never ovary). The first round's candidates are shown on the same development criterion for comparison.

Search candidate (frozen masked features)	Development components met	Losses	Eligible	Registered in the second round
backdoor adjustment	22 of 29	6	no (flips the shortcut)	–
backdoor adjustment, full image	22 of 29	1	yes	–
constraint + backdoor	23 of 29	4	yes	–
constraint + backdoor, full image	23 of 29	1	yes	–
product of experts	28 of 29	1	no (flips the shortcut)	–
group logit adjustment	28 of 29	1	no (flips the shortcut)	–
moment matching	26 of 29	1	yes	–
conditional adversarial debiasing	27 of 29	0	yes	1st
Correct-n-Contrast	20 of 29	3	yes	–
counterfactual contrastive	19 of 29	4	yes	–
V-REx	28 of 29	0	yes	3rd
IRMv1	28 of 29	0	yes	2nd
first round, `mask_cmc` (reference)	27 of 29	1	yes	–
first round, `mask_bal` (reference)	28 of 29	0	yes	–

Table 29: Second registered round, on new confirmation data (seeds 9101–9505): no candidate dominates masking. The first round's two best candidates, refitted with their frozen recipes, are descriptive references on the same data.

Candidate	Components met	All of D2 met	Losses	Role
conditional adversarial debiasing (`mask_condadv`)	27 of 42	no	dermoscopy, MSK held out (D1), ISIC 2019 → 2020 (D1), ovary (held out) (D3), ovary (held out) (D3), ovary (held out) (D3), thyroid+capsule+ovary+isic (D3)	tested, not rejected
IRMv1 (`mask_irm`)	41 of 42	yes	ISIC 2019 → 2020 (D1)	not tested in the sequence
V-REx (`mask_vrex`)	40 of 42	yes	ISIC 2019 → 2020 (D1)	not tested in the sequence
class-conditional mean constraint (`mask_cmc`)	41 of 42	yes	ISIC 2019 → 2020 (D1)	reference (first round)
partial reweighting (`mask_bal`)	38 of 42	yes	thyroid split (D1), dermoscopy, MSK held out (D1), ISIC 2019 → 2020 (D1)	reference (first round)

Table 30: Second round: gains over masking where masking fails (Trap A min(reversed, correlated); conflicting pairs), crossed 95% CIs.

Where masking fails	IRMv1	V-REx	`mask_cmc` (ref.)	`mask_bal` (ref.)
thyroid	+0.229 [+0.205, +0.255]	+0.322 [+0.303, +0.343]	+0.333 [+0.314, +0.353]	+0.393 [+0.363, +0.414]
capsule	+0.030 [+0.008, +0.053]	+0.156 [+0.135, +0.181]	+0.143 [+0.118, +0.172]	+0.164 [+0.140, +0.189]
ovary (held out)	+0.057 [+0.037, +0.080]	+0.081 [+0.055, +0.108]	+0.094 [+0.066, +0.122]	+0.134 [+0.101, +0.170]
dermoscopy hair	+0.120 [+0.096, +0.143]	+0.193 [+0.170, +0.216]	+0.205 [+0.181, +0.229]	+0.218 [+0.194, +0.245]
conflicting pairs, thyroid split	+0.049 [+0.030, +0.073]	+0.088 [+0.061, +0.114]	+0.062 [+0.030, +0.088]	+0.046 [+0.030, +0.063]
conflicting pairs, dermoscopy, BCN held out	+0.017 [+0.013, +0.022]	+0.022 [+0.020, +0.025]	+0.037 [+0.034, +0.039]	+0.028 [+0.021, +0.035]
conflicting pairs, dermoscopy, MSK held out	+0.036 [+0.025, +0.048]	+0.059 [+0.050, +0.067]	+0.055 [+0.026, +0.078]	+0.049 [+0.038, +0.060]
conflicting pairs, ISIC 2019 → 2020	+0.028 [+0.022, +0.035]	+0.043 [+0.037, +0.050]	+0.056 [+0.026, +0.079]	+0.039 [+0.030, +0.048]

Table 31: ISIC 2019 → 2020, second round: the all-pairs difference to masking split exactly into the contributions of the pair types (share of pairs × difference). Every arm, including the unmasked model (ERM), loses mostly on the same-artifact pairs, i.e. disease ranking, not only on the easy pairs.

Arm	Easy pairs	Conflicting pairs	Same-artifact pairs	All pairs
erm	-0.005	0.002	-0.010	-0.014
mask_bal	-0.007	0.002	-0.010	-0.015
mask_cmc	-0.011	0.003	-0.007	-0.015
mask_condadv	-0.006	0.000	-0.024	-0.029
mask_irm	-0.005	0.002	-0.005	-0.009
mask_vrex	-0.009	0.002	-0.006	-0.012

S18. Cohorts and further results in detail

Cohorts in detail

Controlled dermoscopy cohort. ISIC 2018 Task 1/2 images with manual lesion masks and melanoma labels from the union of the ISIC 2016 and 2017 tables (three images with conflicting labels dropped); perceptual-hash (Hamming ≤ 8) and lesion grouping; 2,437 images for which a 65×19 px ruler (0.46% of a 518×518 image) can be placed at 0/25/50/75/100% overlap with the lesion within 10 percentage points (train/val/test 1,849/371/371 before common support).

Real hair cohort (ISIC 2019). All 25,331 ISIC 2019 training images; hair/ruler masks from the archive of Wegley et al. [13]; lesion masks manual for HAM10000 [11] and predicted elsewhere by a U-Net trained on ISIC 2018 Task 1. Two traps share one hair-free group (≤ 30 hair pixels at native resolution): Trap A contains hair with overlap $r = |\text{hair} \cap \text{lesion}|/|\text{hair}| \geq 0.5$, Trap B $r < 0.1$. Cells are subsampled to the hair-free source mix (HAM/BCN/MSK) within each label, giving Trap A 704/157/2,738/989 and Trap B 704/157/4,327/930 images (hair-free benign/melanoma, hair benign/melanoma); five seeds of five-fold group-safe cross-validation (leakage groups: lesion identifiers and pHash); 172 reversed-test melanomas per trap and seed. The 30-pixel threshold is sensitive: independent labels show that it marks hair on many images that experts call hair-free (Robustness paragraphs below).

Thyroid ultrasound. TN3K images [4] with TNCD benign/malignant labels from cytological biopsy [5] (3,493 images of 2,421 patients; 1,210 malignant) and manual nodule masks; the official train/test split (2,879/614 images) is patient-disjoint [5]. Sonographer calipers were segmented by a frozen rule-based detector (dotted measurement lines and '+' ends; audits below and in S4). Trap A: calipers with $r \geq 0.5$ (1,150 images); Trap B: $r < 0.1$ (326); caliper-free: 1,801. Patient identifiers are not published, so within the pooled images leakage groups are pHash chains (stricter embedding-based groups below).

Ovarian tumour ultrasound. MMOTU OTU_2d [15] (CC BY 4.0): 1,202 images with tumour masks, labelled benign cystic (chocolate cyst, serous cystadenoma, simple cyst) versus lesions with solid components (teratoma, theca-cell tumour, mucinous cystadenoma, high-grade serous carcinoma). The thyroid detector is applied unchanged: Trap A 341/276, Trap B 91/103, caliper-free 170/178 (benign-cystic/solid).

Capsule endoscopy. SEE-AI frames [14] (CC BY 4.0) containing only erosions (3,550) or only polyp-like lesions (1,931); the ROI is the union of the expert lesion boxes. Contamination (debris, bubbles, bile) is segmented by a linear probe on DINOv2 patch tokens trained on 2,160 expert-masked frames (held-out pixel accuracy 0.92, IoU 0.62; expert masks are used where available). Debris-free: $< 3\%$ of the frame; Trap A: debris $\geq 10\%$ with $r \geq 0.5$; Trap B: $r < 0.1$. Leakage groups: blocks of 100 consecutive frame numbers plus pHash duplicates.

Chest radiography. NIH ChestX-ray14 [12]; lung ROIs from the TorchXRayVision PSPNet [3]. *Controlled cohort:* 1,906 pneumothorax images without a chest drain (NEATX annotations, 1) and 5,718 *No Finding* images from other patients; a 100×12 px radio-opaque tube is placed at controlled overlap with the lungs (7,610 images in common support). *Real drains* (in-ROI only): 29,687 images; drains are labelled by NEATX [1] for pneumothorax-positive images and by a drain detector (precision 0.95) for negatives; environments are built within view position (AP/PA); five patient-grouped folds.

Further results

Masking implementation and the real overlap (pre-registered). Mean-colour fill is one way to mask. Black fill, a blurred background (16-px Gaussian), a crop to the lesion box and a crop of the masked image all give a positive crossover in all four cohorts (16 of 16, Holm $p \leq 0.010$), from $+0.091$ $[+0.064, +0.117]$ for the dermoscopy box crop to $+0.367$ $[+0.318, +0.416]$ for black fill on capsule frames; implementations that keep part of the out-of-ROI content give weaker effects where that content still carries the artifact (blur for debris and ovarian calipers, the box crop for hair and debris). The two traps are the ends of a continuous dependence: with all artifact-bearing images binned by their real overlap r, the mask gain falls with r in every cohort (slopes: hair -0.235 $[-0.276, -0.194]$, thyroid -0.250 $[-0.303, -0.197]$, capsule -0.489 $[-0.539, -0.438]$, ovary -0.205 $[-0.300, -0.109]$), and for hair it turns from a gain ($+0.124$ $[+0.085, +0.162]$ at $r < 0.1$) into a harm (-0.086 $[-0.119, -0.056]$ at $r \geq 0.75$), crossing zero near $r = 0.40$ (S16).

Replication across encoders, training regimes and foundation models. The crossover replicates with MedSigLIP (thyroid $+0.195$ $[+0.153, +0.238]$, capsule $+0.335$ $[+0.286, +0.383]$, ovary $+0.178$ $[+0.109, +0.248]$), a CNN (ConvNeXt-Base: thyroid $+0.275$ $[+0.231, +0.320]$, capsule $+0.320$ $[+0.268, +0.372]$) and DINOv2 at three sizes (positive in all nine cohort–size combinations; S7), and with end-to-end fine-tuning (crossover of a fine-tuned ResNet-50: hair $+0.143$ $[+0.054, +0.232]$; capsule $+0.606$ $[+0.495, +0.716]$; thyroid ViT-S $+0.122$ $[+0.039, +0.211]$, ResNet-50 $+0.063$ $[-0.022, +0.144]$). The fine-tuned ovary network did not learn the task (ERM clean AUROC 0.60 and 0.50 in Trap A and B, 15 clusters), so it can neither show nor refute a shortcut effect and ovary is not evaluable for this check; the replication holds in all three evaluable cohorts. Without any task training, the zero-shot scores of vision–language models are already swayed by artifacts, and masking changes them as the law predicts (MedSigLIP crossovers: thyroid $+0.103$ $[+0.064, +0.143]$, capsule $+0.045$ $[+0.001, +0.090]$, ovary $+0.181$ $[+0.103, +0.261]$; DermLIP on hair $+0.040$ $[+0.021, +0.058]$); MedSigLIP's zero-shot thyroid diagnosis itself is at chance (clean AUROC 0.45), so we use these scores only to show that the effect is a property of the representation.

The one encoder without a hair gap. With the dermatology foundation model DermLIP and a trained linear head, masking harms at *both* hair locations (Trap A -0.104 $[-0.143, -0.067]$, Trap B -0.090 $[-0.135, -0.046]$) and the hair crossover vanishes ($+0.014$ $[-0.037, +0.063]$), although the controlled ruler sweep reverses with DermLIP as with DINOv2 ($+0.161$ at no overlap, -0.263 to -0.309 at 25–100%). Two measurements explain it. First, with DermLIP masking costs disease signal wherever the hair lies (clean AUROC -0.066 $[-0.089, -0.043]$ in Trap A and -0.063 $[-0.094, -0.033]$ in Trap B): this encoder, trained on whole dermatology images, loses more from a masked background than DINOv2. Second, the pixel part of its hair shortcut is small in both traps (oracle hair inpainting $+0.033$ $[+0.025, +0.042]$ and $+0.040$ $[+0.015, +0.064]$), whereas balancing, which also removes correlates, gains $+0.146$ and $+0.110$: masking removes little of the shortcut at either location, so there is no gap to find. The linear-Gaussian account fitted to DermLIP's clean and correlated AUROC anticipates this (predicted crossover $+0.013$, observed $+0.014$; main-text theory section). The law describes what masking can remove; when the shortcut is carried by correlates and the encoder is fragile to masking, masking removes little at either location.

Operating points: registration status and capsule. Registration status for this frozen model: a pre-registered test inside a post hoc subgroup. The subgroup (the 78 malignant nodules of the official test split that carry an in-ROI caliper; the same 78 in every seed) was defined after the first unaltered-data analysis, where its AUROC harm was found; the operating-point test in it was registered before the operating points were computed, and for the fine-tuned model below the whole test was registered before any result. The pre-registered decision rule (loss at the maximal-balanced-accuracy threshold and at ≥ 3 of the four other operating points) is met. On capsule endoscopy, where masking improves AUROC overall, it lowers the sensitivity for erosions carrying debris on the lesion at the two high-sensitivity operating points (S9).

A fine-tuned model near the benchmark (pre-registered). To test whether these are properties of a weak model, we compared twelve fine-tuning recipes on validation images only, committed the choice (ConvNeXt-T pretrained on ImageNet-22k, learning rate 10^{-4}, 8 epochs, 224 px) and then trained it with five seeds. Its test AUROC is 0.769, 0.004 below the lowest published cross-entropy baseline on this split (0.773; 5) and above the frozen probe (0.735). All four registered tests are supported (Holm): masking lowers the AUROC on shortcut-conflicting pairs by -0.102 $[-0.178, -0.026]$; it lowers sensitivity at all four of OP2–OP5; at OP2 it lowers sensitivity among the 78 malignant nodules with an in-ROI caliper from 0.744 to 0.223 (-0.521 $[-0.642, -0.382]$), a subgroup registered before any result of this model existed; and its trap crossover is $+0.187$ $[+0.111, +0.268]$. Group balancing recovers the conflicting pairs ($+0.088$ $[+0.019, +0.158]$ over masking).

Is the loss a threshold artefact? (pre-registered). Re-tuning each model's threshold on the test set itself—the best that recalibration at a deployment site can achieve—makes the overall sensitivity of the two models equal at a test specificity of 0.80 (-0.017 $[-0.124, +0.070]$ for the fine-tuned model), but the conflicting subgroup still loses: from 0.544 to 0.390 (-0.154 $[-0.302, -0.023]$) for the fine-tuned model and from 0.505 to 0.331 (-0.174 $[-0.313, -0.012]$) for the frozen probe (Holm $p = 0.019$ each), while the malignant nodules whose caliper points the right way tend to gain. About 30% of the loss at validation-fixed thresholds therefore cannot be removed by any threshold; the rest follows from transporting thresholds from validation to test data, as deployment does. At a matched specificity of 0.90 and a matched sensitivity of 0.80 the direction is the same, with intervals that include zero (S16).

New patients at natural prevalence (pre-registered). Trained on all ISIC 2019 images and tested on ISIC 2020 without any resampling, masking lowers the AUROC on shortcut-conflicting pairs: -0.026 $[-0.052, -0.000]$ under the registered near-duplicate rule, which proved too loose (it removed most look-alike images of other patients), and -0.037 $[-0.061, -0.012]$ under a label-free calibrated rule registered as a sensitivity analysis after that result (32,952 images). The overall AUROC is nearly unchanged ($+0.016$ $[-0.000, +0.033]$), and group balancing recovers the conflicting pairs ($+0.072$ $[+0.048, +0.097]$ over masking). No sensitivity loss at a fixed operating point was detectable, and the cleanest in-lesion comparison (53 hair-free melanomas) was inconclusive (-0.005 $[-0.057, +0.048]$): the external harm combines the removal of hair beside melanomas with the retention of hair on benign lesions (S16).

Calibration. For the held-out BCN hospital masking worsens both Brier score and expected calibration error (Table 32); for HAM, MSK and capsule it improves every metric, in line with the AUROC results.

Robustness to artifact-label definitions. Artifact masks for calipers come from a detector (visual audit: 95% of thyroid detections lie on real calipers; 92–97% of caliper-free images contain no visible caliper). Re-running the thyroid and ovary analyses with large-marker-only (≥ 50 px) and strict-location ($r \geq 0.7$ / $r < 0.05$)

Table 32: Clinical secondary metrics on the unaltered test sets (main text, unaltered test data): mean (SD) over training seeds. Sensitivity and specificity at the operating point of maximal balanced accuracy fixed on validation data (OP1; other operating points with CIs in the main-text operating-point table and S9); ECE with 10 equal-width bins; AUPRC baseline = prevalence. Rebuilt run (S18).

Test set	Arm	AUROC	AUPRC	Brier	ECE	Sens	Spec
Thyroid (official split) ($\pi = 0.38$)	ERM	0.735 (0.006)	0.604 (0.006)	0.209 (0.002)	0.088 (0.003)	0.699 (0.045)	0.653 (0.037)
	mask	0.731 (0.005)	0.629 (0.005)	0.215 (0.006)	0.119 (0.025)	0.424 (0.048)	0.856 (0.021)
	U-MtE	0.735 (0.007)	0.630 (0.007)	0.206 (0.003)	0.087 (0.007)	0.419 (0.043)	0.853 (0.020)
	balanced	0.714 (0.005)	0.580 (0.003)	0.220 (0.003)	0.115 (0.006)	0.707 (0.070)	0.612 (0.069)
	U-MtE$_{\text{bal}}$	0.726 (0.006)	0.620 (0.012)	0.207 (0.004)	0.072 (0.024)	0.408 (0.053)	0.851 (0.040)
ISIC, BCN held out ($\pi = 0.23$)	ERM	0.786 (0.003)	0.587 (0.003)	0.238 (0.006)	0.279 (0.011)	0.842 (0.038)	0.521 (0.075)
	mask	0.761 (0.004)	0.564 (0.010)	0.287 (0.007)	0.330 (0.010)	0.842 (0.016)	0.443 (0.033)
	U-MtE	0.767 (0.003)	0.570 (0.007)	0.280 (0.008)	0.327 (0.013)	0.853 (0.011)	0.443 (0.020)
	balanced	0.782 (0.006)	0.587 (0.010)	0.229 (0.006)	0.260 (0.012)	0.835 (0.033)	0.509 (0.071)
	U-MtE$_{\text{bal}}$	0.771 (0.004)	0.567 (0.008)	0.282 (0.010)	0.329 (0.016)	0.869 (0.026)	0.418 (0.058)
ISIC, MSK held out ($\pi = 0.19$)	ERM	0.770 (0.008)	0.477 (0.012)	0.171 (0.007)	0.160 (0.019)	0.600 (0.062)	0.790 (0.048)
	mask	0.785 (0.003)	0.497 (0.006)	0.149 (0.003)	0.120 (0.011)	0.587 (0.060)	0.823 (0.043)
	U-MtE	0.778 (0.002)	0.488 (0.005)	0.154 (0.003)	0.134 (0.008)	0.575 (0.070)	0.818 (0.048)
	balanced	0.767 (0.008)	0.464 (0.014)	0.175 (0.008)	0.165 (0.019)	0.628 (0.043)	0.759 (0.043)
	U-MtE$_{\text{bal}}$	0.770 (0.004)	0.470 (0.006)	0.160 (0.004)	0.141 (0.010)	0.592 (0.050)	0.795 (0.042)
ISIC, HAM held out ($\pi = 0.11$)	ERM	0.752 (0.006)	0.355 (0.006)	0.161 (0.007)	0.233 (0.012)	0.575 (0.084)	0.789 (0.062)
	mask	0.782 (0.004)	0.396 (0.011)	0.138 (0.005)	0.199 (0.011)	0.569 (0.075)	0.831 (0.056)
	U-MtE	0.795 (0.003)	0.402 (0.011)	0.139 (0.006)	0.208 (0.012)	0.606 (0.067)	0.819 (0.046)
	balanced	0.751 (0.016)	0.350 (0.018)	0.154 (0.013)	0.225 (0.014)	0.495 (0.063)	0.845 (0.046)
	U-MtE$_{\text{bal}}$	0.800 (0.008)	0.399 (0.012)	0.135 (0.004)	0.206 (0.015)	0.596 (0.062)	0.828 (0.045)
Capsule (held-out frames) ($\pi = 0.56$)	ERM	0.889 (0.019)	0.897 (0.046)	0.130 (0.016)	0.042 (0.011)	0.770 (0.062)	0.844 (0.055)
	mask	0.967 (0.011)	0.971 (0.014)	0.069 (0.013)	0.029 (0.011)	0.856 (0.034)	0.933 (0.025)
	U-MtE	0.962 (0.009)	0.967 (0.013)	0.081 (0.017)	0.065 (0.045)	0.865 (0.036)	0.904 (0.045)
	balanced	0.881 (0.024)	0.889 (0.053)	0.136 (0.017)	0.051 (0.009)	0.757 (0.086)	0.838 (0.058)
	U-MtE$_{\text{bal}}$	0.959 (0.008)	0.964 (0.012)	0.079 (0.007)	0.045 (0.012)	0.866 (0.050)	0.897 (0.051)

definitions leaves the crossover (thyroid +0.215 to +0.262; ovary +0.170 to +0.195) and the U-MtE$_{\text{protect}}$ gain over masking (thyroid +0.217 to +0.282; ovary +0.043 to +0.067) significant in every variant; both grow under stricter definitions, as expected if detector errors dilute rather than create the effects.

Artifact labels checked without a clinician (pre-registered). No clinician was available to audit the labels, so we compared them with every independent source we could find, with coverage gates and without the images on which our own labellers were trained (S16). *Dermoscopy.* Where hair is present our labels place it correctly: our lesion masks agree with the IMA++ multi-annotator masks (241 images; Dice 0.851 [0.830, 0.870]; trap class agreement 84%), and against independent hair masks (485 images) the hair lies on the other side of the lesion boundary in 1 of 264 trap images. But at our threshold (> 30 pixels) the published hair masks mark hair on 93% of the 8,864 images that the expert-reviewed DermArtifactDB labels call hair-free ($\kappa = 0.076$), more often for melanomas. Removing every image an independent source contradicts (42% of each trap) raises the hair crossover from +0.147 [+0.111, +0.185] to +0.199 [+0.164, +0.233]: the label noise diluted the effect. *Calipers.* Two independent labellers—an ultrasound annotation detector (BUSClean)

and a medical vision–language model (MedGemma, run locally with fixed prompts and never shown our masks)—both pass a pre-registered informativeness rule on thyroid ($\kappa \geq 0.40$) but err in opposite directions, so the registered rule that removes an image whenever either disagrees left too few out-of-ROI caliper images to re-run the thyroid traps. Where the two agree with each other (1,865 of 3,277 images) they agree with our cell for 94%, and without the images both contradict the thyroid crossover is $+0.257$ $[+0.209, +0.304]$ (exploratory; $+0.239$ on our labels); the subgroup loss of the main-text section on unaltered data holds in the 36 nodules whose in-ROI caliper both confirm. On ovarian screens the annotation detector fires on 90% of caliper-free images and was declared invalid, so the ovary and capsule labels could not be checked with public data.

Stricter leakage grouping (pre-registered). TN3K/TNCD, MMOTU and SEE-AI publish no patient or video identifiers, so the main analyses group near-duplicates by perceptual hash (and, for capsule, blocks of 100 consecutive frames). As a pre-registered sensitivity analysis we united these groups with connected components of mean-centred DINOv2 embeddings at the most aggressive cosine threshold that keeps every group below 5% of the cohort (thyroid: 3,455 $\rightarrow$ 3,081 groups, $\tau = 0.80$; ovary: 1,059 $\rightarrow$ 969, $\tau = 0.80$; capsule: 137 $\rightarrow$ 135, $\tau = 0.93$) and re-ran all primary contrasts. Eleven of twelve keep their sign and significance: the location crossover is $+0.235$ $[+0.193, +0.275]$ (thyroid), $+0.157$ $[+0.053, +0.255]$ (ovary) and $+0.353$ $[+0.296, +0.410]$ (capsule), and protected U-MtE still beats masking on thyroid ($+0.251$ $[+0.215, +0.284]$) and ovary ($+0.051$ $[+0.020, +0.079]$). The exception is the small ovary advantage of erasure over augmentation, which loses significance ($+0.024$ $[-0.014, +0.062]$). Embedding grouping removes visually near-identical images of one patient or session; it cannot exclude differently looking images of the same patient.

Rebuild on new hardware (pre-registered). All data were downloaded again and every derived object—caches, perceptual-hash groups, the retrained lesion U-Net (held-out Dice 0.886; HAM10000 agreement 0.895), the caliper scan and the debris probe (IoU 0.625)—was rebuilt on a different GPU generation (pre-registered). Trap membership changes slightly where it depends on a retrained model (hair traps: Trap A 2,655/942 artifact-bearing benign/melanoma images instead of 2,738/989); the thyroid and ovary cells are identical. Every primary crossover reproduces the archived point estimate within the pre-registered tolerance of 0.03 (rebuilt, crossed CIs: dermoscopic hair $+0.147$ $[+0.109, +0.185]$; thyroid $+0.239$ $[+0.196, +0.280]$; ovary $+0.170$ $[+0.087, +0.253]$; capsule $+0.377$ $[+0.326, +0.425]$).

Limitations in detail. (i) *Artifact labels.* Calipers are segmented by a rule-based detector and capsule debris by a patch-token probe (IoU 0.62 against expert masks); hair masks are published semi-automatic masks. No clinician audited them. Independent sources confirm where hair lies, but our hair-presence threshold over-calls hair, with errors that differ by diagnosis; the thyroid caliper labels agree with two independent labellers where those agree with each other, but the registered cleaning rule could not be tested

for thyroid, and the ovary and capsule labels remain unchecked (Robustness paragraphs above). A blinded rating of 240 images by the first author, a non-clinician, is registered and not yet done. Label errors that do not depend on the diagnosis dilute the constructed correlation, and removing contradicted images or using stricter label definitions strengthens rather than weakens the results (Robustness paragraphs above); the transplant experiment does not depend on the artifact-free group being perfectly clean because the transplanted instance is identical at both locations. (ii) *Patient identifiers.* TN3K/TNCD, MMOTU and SEE-AI publish no patient or video identifiers for the pooled images, so the trap splits group near-duplicates by perceptual hash and frame blocks (a stricter embedding grouping leaves 11 of 12 primary estimates unchanged); the official TN3K/TNCD test split used for the unaltered-data analysis is patient-disjoint. The one open cohort we found with patient identifiers, pathology labels and real calipers (ThyUS2Path; 842 patients) has no nodule masks and uses a scanner interface on which our frozen caliper detector fires on interface elements; external validation on it requires a transferred nodule segmentation and a re-validated detector and is left for future work. For dermoscopy, external validation on new patients was possible and was run under a pre-registered cell-count gate (ISIC 2020, patient-disjoint folds, hair masks from a segmenter trained only on ISIC 2019): the crossover replicates ($+0.250$ $[+0.174, +0.327]$), and at natural prevalence masking lowers the AUROC on conflicting pairs (main text, unaltered test data). External validation is limited to dermoscopy, and on ISIC 2020 no operating-point effect was detectable. (iii) *Constructed shifts.* Reversed environments are constructed by subsampling; real deployment shifts are less extreme, which is why we report the unaltered test distributions separately. (iv) *Absolute performance.* Frozen encoders with linear heads are below the best published fine-tuned models on the one cohort with a directly comparable benchmark (official TNCD split: 0.735 AUROC versus 0.773–0.784 for fine-tuned CNNs with cross-entropy and up to 0.799 with curriculum learning, 5). A fine-tuned ConvNeXt-T reaches 0.769, near but below the lowest published baseline, and shows every registered harm (main text, unaltered test data). The law replicates with end-to-end fine-tuned networks in three of four cohorts; on ovarian ultrasound the fine-tuned ResNet-50 barely learns the task and shows no significant crossover (Replication paragraph above). The other cohorts' binary tasks have no published benchmark on the same split (Supplement S10). (v) *Scope.* Chest radiography is a boundary case: the radiograph encoder ignored extra-pulmonary tubes, the crossover that a second encoder shows rests on a device contrast confounded with other tubes, and real drains carry the shortcut through correlates. The linear-Gaussian account is fitted per model and cannot anticipate a shortcut carried by correlates that the clean and correlated environments do not expose. (vi) *Uncertainty and balance.* The original per-seed bootstrap understated CIs where seeds share test images. All model-comparison intervals were therefore recomputed with the crossed bootstrap from regenerated runs (main-text protocol section); claims whose interval then included zero were rewritten, and a few secondary analyses from runs that could not be regenerated are given as point estimates only. Matching left within-label standardised differences of up to 0.19 for some covariates. (vii) *Transplant.* Pasted patterns, including neutral tissue,

become learnable shortcuts; the transplant establishes location dependence on fixed images, not the size of real artifacts' effects in their natural context (for hair, whose natural shortcut is carried mostly by correlates, the difference is large). (viii) *Pre-registration.* Every registration is committed before its outcome results (registry with commit hashes and times in Supplement S1); commit times are set by the committing machine, so only the registrations that were pushed to GitHub before their runs (the second review round and every later round) carry an external time stamp, and the 16-test primary family was assembled after the individual results. Two later analyses are sensitivity analyses registered after the result they examine (the calibrated duplicate rule, the cleaner hair groups), and a few are exploratory (the consensus caliper labels). (ix) *Operating points.* Most of the thyroid sensitivity loss at validation-fixed thresholds is threshold transport; the part that survives re-tuning has a wide interval (-0.154 $[-0.302, -0.023]$). (x) *Remedies.* No remedy dominates masking: in two registered rounds the best candidates fail on one cell, ISIC 2020 overall, by 0.01–0.02 AUROC; the dominance tests use frozen DINOv2 features, and the one fine-tuned candidate (a paste-based model) was not a fair test because its pasting could not balance the traps.

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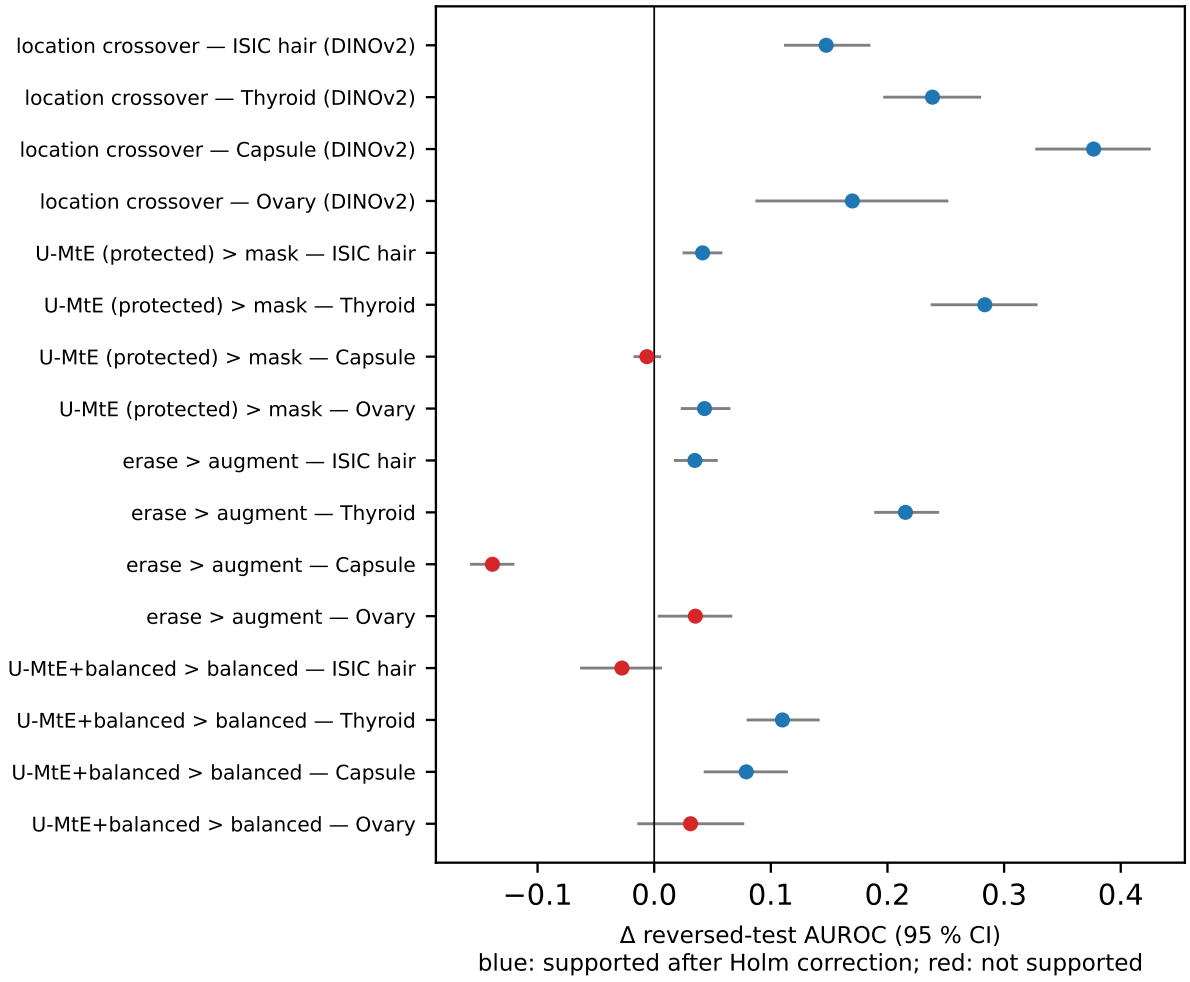


Figure 1: The 16 pre-registered primary tests with Holm correction (blue: supported; red: not supported). Generated by `scripts/analysis/primary_claims.py` and `scripts/make_figures.py`.

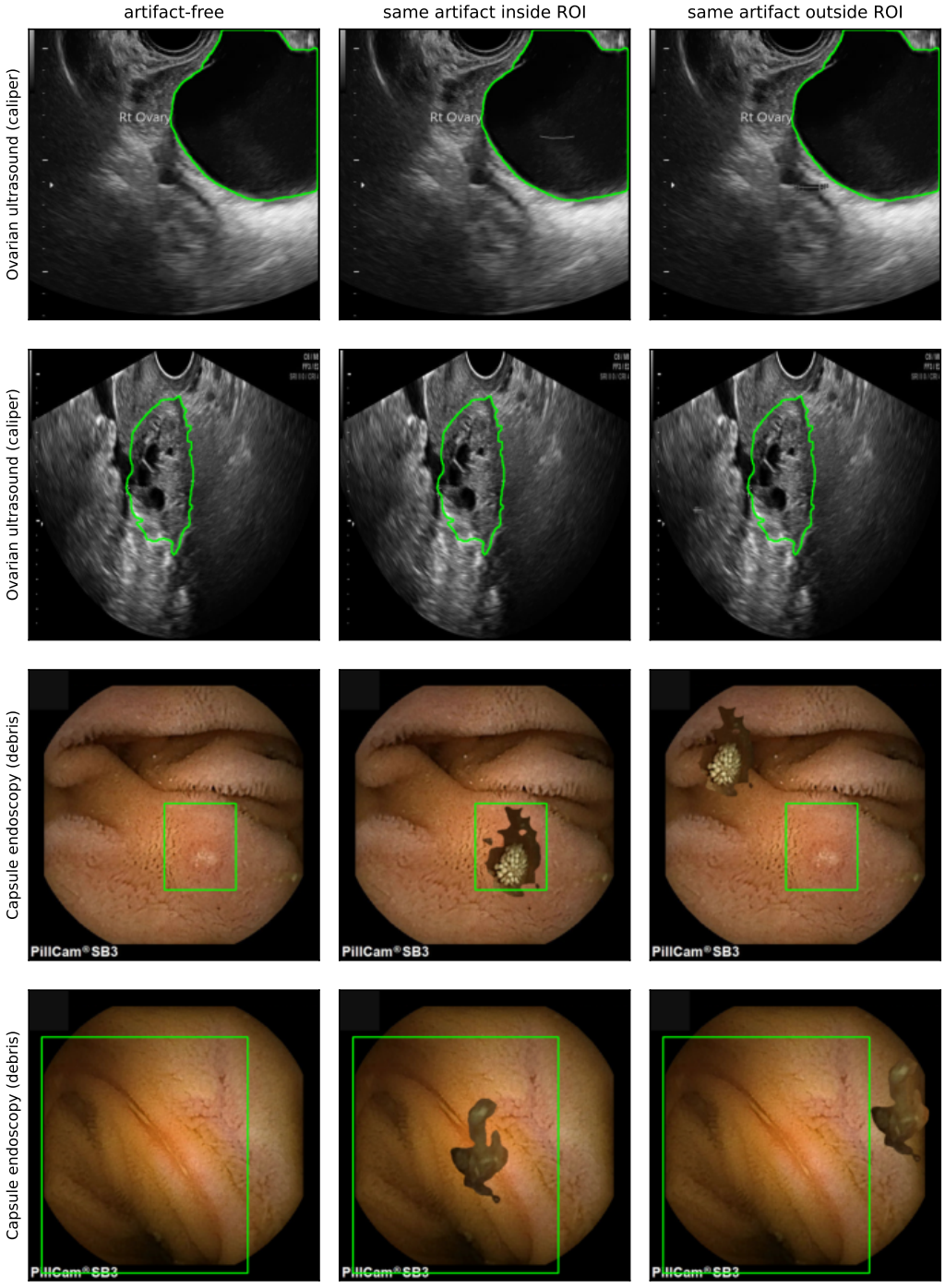


Figure 2: Real-artifact transplant, openly licensed cohorts (green: ROI). Each row shows one artifact-free image and the same real artifact instance—cut from another image of the cohort with its mask—pasted inside and outside the ROI. Instances carry a thin rim of donor background where the mask extends beyond the artifact; the rim is identical at both locations. Generated by `scripts/make_transplant_figure.py`.